

Project 2 Implementation Stage — transcript

- **Project:** ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint Calculator
- **Repository:** <https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator>
- **Branch:** project-2 (HEAD at start of session: ba11478; at save: 5922577, pushed to origin/project-2)
- **Date:** 2026-09-26
- **Learner:** RG Sureshkumar
- **Agent:** Claude Code (Opus 5.5)
- **Stage command:** start implementation (instructions in plan.md, Commands section)
- **Outcome:** All six checkpoints of the approved plan were built, checked against the specification's acceptance checks in the footprint preview, and committed at the learner's direction (6b57f78, 397e7eb, 3210da9, c6624c1, e793423, 87fdd88). Implementation-stage decisions made by the learner (logged prompts follow "Use my log"; console streaming central value, router hours, and log window for feature 3; the S51 phone-water mismatch; removal of the team view's shower anchor) are recorded under Revisions in spec.md and plan.md. After a first save of this transcript (committed as ab1a9a0), the learner had the branch pushed and directed the agent to tick the six checkpoint verification items in plan.md (committed and pushed as 5922577); this file replaces that first save and covers the whole chat. The four Verification items in plan.md remain open.

Note on redaction. Sanitised following transcripts/README.md: the author's email address, macOS home path and username, and temporary/session paths are replaced with placeholders. No messages were removed or reworded.

Note on tool calls. Each tool call is shown as a one-line summary in square brackets, followed by its output truncated to 400 characters, except the learner's answers to the agent's multiple-choice questions, which are shown in full because they record decisions. Screenshots appear as [screenshot]. The agent's internal reasoning is not included. The complete local session log is kept locally and not committed. Automatic system notices shown to the agent during the session (such as file-change notices) are omitted.

Note on the final turn. The last learner turn (save transcript) is followed by the tool calls that generated this file; the agent's closing confirmation was written after the file was generated and so is not included.

1. Learner (RG Sureshkumar)

start implementation

Agent

[Tool: Bash — List project files and plan headings]

```
AGENTS.md
CLAUDE.md
LICENSE
README.md
assets
brief.md
index.html
plan.md
research.md
spec.md
transcripts

./assets:
css
js

./assets/css:
base.css
features.css

./assets/js:
app.js
budget.js
calculator.js
core.js
data.js
nav.js
store.js
team-data.js
team.js
tracker.js

./transcripts:
README.md
debrief-2026-09-12_000739.md
debrief-2026-09-12_000739.pdf
plan-2026-09-26_194158.md
plan-2026-09-26_1941 ... [truncated]
```

[Tool: Bash — Read the implementation plan]

```
# Implementation Plan

> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST
REVIEW AND APPROVE ITS CONTENT.

Purpose of this file: Turn the approved specification into ordered, updatable
implementation and verification work.

## Instructions for the user

Preserve the approved requirements and verify the completed work. Direct priorities,
scope, and meaningful checkpoi ... [truncated]
```

[Tool: Bash — Read brief and specification]

```
<persisted-output>
Output too large (50.1KB). Full output saved to: <session-log>

Preview (first 2KB):
# Project 2 Brief

> EDITING DIRECTIVE: DO NOT EDIT THIS FILE ... [truncated]
```

[Tool: Bash — Size up scripts and read core and app]

```
31 assets/js/app.js
    296 assets/js/budget.js
    422 assets/js/calculator.js
    200 assets/js/core.js
    169 assets/js/data.js
    159 assets/js/nav.js
     82 assets/js/store.js
    107 assets/js/team-data.js
    240 assets/js/team.js
    375 assets/js/tracker.js
    484 index.html
    2565 total

/**
 * core.js – shared state, formatting, and the carbon / water math.
 *
 * Base uni ... [truncated]
```

[Tool: Read — bwsmoetbt.txt]

```
1  # Project 2 Brief
2
3  > EDITING DIRECTIVE: DO NOT EDIT THIS FILE. IT IS THE AUTHORITATIVE PROJECT FRAME.
4
5  Roles in the project documents: The user is the person directing and working with
the AI agent. The agent supports the user's work but does not make or approve decisions
on the user's behalf. Employees are the intended audience for the completed calculator,
and the named reference profi ... [truncated]
```

[Tool: Bash — Read store and tracker; find AI-total call sites]

```
/**
 * store.js – namespaced localStorage with safe fallbacks.
 *
 * Every read and write is wrapped: a private window, cleared site data, or a
 * browser set to block storage all throw on access rather than returning
 * empty, and the page has to render correctly anyway. When storage is
 * unavailable we fall back to an in-memory map so a session still works for as
 * long as the tab is open.
 */ ... [truncated]
```

I've read the instructions, brief, spec, and plan, and the code the first checkpoint touches. Here's where things stand.

Where we are - All three approval gates in `plan.md` are ticked with your approval dates, so implementation can start. - There are 6 checkpoints. After each one, I run its acceptance checks in the footprint preview. Then you verify, and I commit only when you ask.

Checkpoint 1: one shared place to compute the AI total. This step should change no figures. -

Current state: the AI total is computed in several separate places: - `aiDailyTriple`, `aiDaily` and `aiDailyEnergy` in `core.js`, which the calculator, the nav total and the team view read - `sumEntries` and `trackerDayTotal` in `tracker.js`, which the tracker's running total, its history and the budget read -

Change: - Tracker log entries get a `k` (kind) field. Saved entries without one are read as prompts. - One shared function returns the AI total (central, low and high) plus an energy figure, covering the rows and today's logged images and sessions. - Every call site above switches to that function. For now it still counts only prompts, because images and sessions don't exist until checkpoints 2 and 3. - **Old "agent" entries** keep counting in this step. The plan removes them in checkpoint 3. - **Before/after check:** 1. In the preview, I record every figure on the page: headline, running line, charts, tracker total, 7-day history, budget, nav and team. I do this for the example day, then for a sample log I add to the preview browser's own storage (not your real browser). 2. After the change, I repeat the same readings and compare them value by value.

My question: should I start checkpoint 1 now, using that before/after check? Or do you want to change the scope or the check first?

2. Learner (RG Sureshkumar)

start checkpoint 1 now

Agent

[Tool: Bash — Read calculator results code and other call sites]

```
const rm = el('button', 'aipf-rowdel', 'x');
rm.type = 'button'; rm.setAttribute('aria-label', 'Remove this row');
rm.addEventListener('click', () => {
  state.rows = state.rows.filter((x) => x.uid !== row.uid);
  renderRows(); AIPF.emitUpdate();
});
r.appendChild(rm);
wrapEl.appendChild(r);
wrapEl.appendChild(el('div', 'aipf-row-meta'));
wrap. ... [truncated]
```

[Tool: Bash — Read the rest of the report builder and script order]

```
wDay += r.count * wt[0]; wMin += r.count * wt[1]; wMax += r.count * wt[2];
rowsHtml += '<tr><td>' + escXml(md.name) + '</td><td>' +
escXml(AIPF.sizeLabel(r.model, AIPF.getSize(r.size))) + '</td><td class="n">' + r.count
```

```
+ '</td><td class="n">' + AIPF.fmtCarbon(d) + '</td><td class="n">' +
AIPF.fmtWater(r.count * wt[0]) + '</td></tr>';
}
const aiYr = aiDay * DAYS, wYr = wDay * DAYS; ... [truncated]
```

Starting checkpoint 1. First I'll take the "before" snapshot in the preview.

[Tool: Browser preview_start — footprint]

```
{
  "serverId": "62b7c849-b64a-4d05-afbf-c81d5f85aed9",
  "port": 4173,
  "name": "footprint",
  "reused": false,
  "previewId": "preview-local_75563973-ee26-49cb-87d9-d976bbd0b65c",
  "tabId": "seed"
}
Server started successfully on port 4173. Opened tab "seed" at http://localhost:4173.
Use tabId "seed" with read_page / computer / navigate; serverId is for preview_stop and
preview_logs only.
```

[Tool: Bash — Read the report-generation function]

```
function generateReport() {
  try {
    const blob = new Blob([reportHTML()], { type: 'text/html' });
    window.open(URL.createObjectURL(blob), '_blank');
  } catch (e) { /* ignore */ }
}
AIPF.reportBarsSVG = reportBarsSVG;
```

[Tool: Bash — Write the before/after snapshot helper script]

```
ok
```

[Tool: Browser javascript_tool — javascript_exec]

```
"reloading"

Tab Context:
- Executed on tabId: seed
- Available tabs:
  • tabId seed: "AI Footprint Calculator — Company Edition" ("http://localhost:4173/")
```

Scenario A (example day, empty log):

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "energy": 51.2619000000000004,
```

```
"len": 17479,  
"report": 21222,  
"triple": [  
  20.847521999999998,  
  14.450146000000004,  
  27.244328  
]  
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
{  
  "days": 5,  
  "driving": true,  
  "energy": 496.1969999999995,  
  "triple": [  
    198.38555999999997,  
    138.18025,  
    258.59088399999996  
  ]  
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
{  
  "energy": 51.261900000000004,  
  "tk": "198 g CO2e",  
  "triple": [  
    20.847521999999998,  
    14.450146000000004,  
    27.244328  
  ],  
  "useLog": false  
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Before-snapshots are saved for three cases: the example day, a seeded log with the log on, and the same log with it off. Now the code changes.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

Now the tracker.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

Now the report, which kept its own running sums:

[Tool: Edit — calculator.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/calculator.js has been updated successfully.

[Tool: Bash — Check leftover variables and syntax of all scripts]

```
293:    const [aiDay, aiMin, aiMax] = AIPF.aiDailyTriple('carbon');
341:    '<p>Across these prompts, my AI use comes to about <strong>' +
AIPF.fmtCarbon(aiDay) + ' per day</strong> (range ' + AIPF.fmtCarbon(aiMin) + ' to ' +
AIPF.fmtCarbon(aiMax) + '), or <strong>' + AIPF.fmtCarbon(aiYr) + ' per year</strong>,
drawing about ' + AIPF.fmtEnergy(eD) + ' of electricity a day.' + fn(1) + fn(2) + ' T ...
[truncated]
```

Water range is still needed — fixing that line:

[Tool: Edit — calculator.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/calculator.js has been updated successfully.

Now the "after" snapshots, compared field by field with the "before" ones:

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: TypeError: Cannot read properties of undefined (reading
'state')
    at window.cp1snap (eval at <anonymous> (:1:2), <anonymous>:13:7)
    at window.cp1cmp (eval at <anonymous> (:1:45), <anonymous>:3:49)
    at <anonymous>:2:17
```

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Probably ran before the page finished loading; retrying.

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "diffs": [],
  "energy": 51.261900000000004,
  "id": "C",
  "triple": [
    20.847521999999998,
    14.450146000000004,
    27.244328
  ]
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser navigate — http://localhost:4173/]

navigated to http://localhost:4173/

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "diffs": [],
  "driving": true,
  "energy": 496.19699999999995,
  "id": "B",
  "triple": [
    198.38555999999997,
    138.18025,
    258.59088399999996
  ],
  "useLog": true
}
```

```
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

"cleared"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser navigate – http://localhost:4173/]

navigated to http://localhost:4173/

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
{
  "diffs": [],
  "energy": 51.261900000000004,
  "extra": {
    "afterUndo": 0,
    "one": 0.32601800000000003,
    "rows": 1,
    "storedKind": "prompt",
    "withExtras": 0.32601800000000003
  },
  "id": "A",
  "triple": [
    20.847521999999998,
    14.450146000000004,
    27.244328
  ]
}
```

Tab Context:

- Executed on tabId: seed

- Available tabs:
 - tabId seed: "AI Footprint Calculator – Com ... [truncated]

All three cases match exactly. Cleaning up the test data and checking for console errors:

[Tool: Browser javascript_tool — javascript_exec]

[]

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 400x304 0.5-scale view; coordinate frame: 800x609.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Page loads cleanly. Updating the plan's checkpoint 1 tasks (agent tasks only; your verification stays open):

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

[Tool: Bash — Summarize changed files]

```
assets/js/calculator.js | 10 +++++-----
assets/js/core.js       | 44 ++++++++++++++++++++++++++++++++++++++-----
assets/js/tracker.js    | 16 ++++++-----
```

```
plan.md | 8 ++++----
4 files changed, 56 insertions(+), 22 deletions(-)
```

Checkpoint 1 is built, and every figure on the page is unchanged. It's ready for you to verify.

What changed - Entry kinds ([core.js](#)): - Log entries can now be one of four kinds: prompt, image, session or digital activity. Only the first three count toward the AI total. - Saved entries with no kind are read as prompts, and new prompts are saved as k: 'prompt'. - `entryTriple` returns an entry's central, low and high values in carbon, water or energy. Images and sessions return zero until checkpoints 2 and 3. - **One AI total:** `aiDailyTriple` in `core.js` is now the only place the AI total is computed. - It adds today's logged images and sessions to the prompt rows, but only when the log is driving the page. - `aiDailyEnergy` now comes from the same function. - Logged prompts reach the total through the rows as before, so they aren't counted twice. - **Tracker** ([tracker.js](#)): the running total, hourly bars, recent list, 7-day history and `trackerDayTotal` (which the budget reads) all use the shared entry costs. Only prompt entries become calculator rows. - **Report** ([calculator.js](#)): it no longer keeps its own sums; its totals and ranges come from the shared function. - The headline, running line, charts, nav total and team view already read the AI total, so they needed no changes.

Before/after check - I compared the page in three cases: 1. the example day with an empty log 2. a sample log driving the page: 7 prompts today, including an old "agent" entry, plus 4 earlier days 3. the same log with "Use my log" turned off - In each case I compared, in both carbon and water: all page text, all tooltips, the AI total and its range, 40 days of tracker totals, daily energy and the full generated report. There were **no differences in any case**. - I also checked that: - a newly logged prompt is saved with its kind - image and digital-activity entries add nothing to the AI total yet and don't become rows - "Undo last" still works - There are no console errors, and I cleared all test data from the preview's storage afterwards.

In `plan.md` I ticked the four agent tasks for checkpoint 1. "User verifies checkpoint 1; commit when asked" is left for you.

To check it yourself: - Open the page and log a few prompts. The tracker total, headline, charts, budget and nav total should behave exactly as before. - Old entries saved before this change should still count. - For the code, run:

```
git diff
```

Nothing is committed yet. Once you've verified this checkpoint, tell me whether to commit it or what to change.

3. Learner (RG Sureshkumar)

commit checkpoint 1

Agent

[Tool: Bash — Commit checkpoint 1 on project-2]

```
project-2
6b57f78 Compute the AI total in one place and give log entries a kind
```

Checkpoint 1 is committed on project-2 as 6b57f78, with the four code files and plan.md. The working tree is clean, and I haven't pushed.

I left "User verifies checkpoint 1; commit when asked" unticked in plan.md, because only you can mark your own verification. If you've checked it and want the box ticked, say so and I'll tick it and record the date.

Next is checkpoint 2, image logging: - The image figures go into data.js with their sources: 0.086, 1.35 and 11.49 Wh per image, and 4.88 L/kWh for water. - The tracker gets an "Image" action with a count. - Image results show a central value with a low-high range, plus the notes on limitations and the methodology text. - Then I run feature 1's acceptance checks 1-6.

Should I start checkpoint 2?

4. Learner (RG Sureshkumar)

start checkpoint 2

Agent

[Tool: Bash — Find image and water sources in research]

```
33:Evidence (figures checked against the source text; see Source assessments S1-S3):
35:- Energy per image varies by more than two orders of magnitude across studies: about
0.09 Wh to 4.1 Wh across 17 open models on a consumer GPU (S3); a median of 1.35 Wh,
mean of 2.9 Wh, and maximum of 11.5 Wh across 2022-23 open models on a data-center GPU
(S1); about 0.32 Wh GPU-only, or about 0.63 Wh with ove ... [truncated]
```

[Tool: Bash — Read source entries and data.js structure]

```
### S1. Luccioni, Jernite & Strubell (2024)

- Citation: Luccioni, A. S., Jernite, Y., & Strubell, E. (2024). Power Hungry
Processing: Watts Driving the Cost of AI Deployment? *Proceedings of the 2024 ACM
Conference on Fairness, Accountability, and Transparency (FAcT '24)*.
<https://arxiv.org/abs/2311.16863>
- Claim or figure: image generation used a median of 1.35 kWh, and a mean of 2.907 kWh
(S ... [truncated]
```

Checking the four source links directly before the figures go into the data:

[Tool: Bash — Check HTTP status of the five source URLs]

```
200 https://arxiv.org/abs/2311.16863
200 https://www.technologyreview.com/2025/05/20/1116327/ai-energy-usage-climate-
footprint-big-tech/
```

```
200 https://www.technologyreview.com/2025/05/20/1116331/ai-energy-demand-methodology/
200 https://arxiv.org/abs/2506.17016
200 https://eta-publications.lbl.gov/sites/default/files/2024-12/lbnl-2024-united-
states-data-center-energy-usage-report_1.pdf
```

All return 200. Spot-checking the exact figures in the source text:

[Tool: Bash — Verify figures in S1, S3, S46 source texts]

```
ugrep: error: error at position 204
0-\xbf][\x80-\xbf]*){60}
      \__exceeds complexity limits

egmind Stable Diffusion 1B (SSD_1B) [ 29 ] , Latent Consistency Model Segmind Stable
Diffusion 1B (LCM_SSD_1B) [ 30 ] , Latent Consistency Model Stable Diffusion XL
(LCM_SDXL) [ 30 ] , Flash Stable Diffusion
ina being the models consuming the highest energy value ( 4.08 × 10 − 3 ... [truncated]
```

[Tool: Bash — Check S1 median and maximum image figures]

```
image generation (1.35 kWh) and, conversely, text generation (0.042 kWh) requiring more
than text classification (0.0023 kWh). For comparison, charging the average smartphone
requires 0.022 kWh of energy ( US Enviro
ugrep: error: error at position 83
xbf][\x80-\xbf]*){0,40}
      \__exceeds complexity limits
Shell cwd was reset to ~/Documents/College/Undergra ... [truncated]
```

[Tool: Bash — Show context around S1's 11.49 figure]

```
% of a full smartphone charge for 1,000 inferences, whereas the least efficient image
generation model uses as much energy as 522 smartphone charges (11.49 kWh), or around
half a charge
--
e. those trained for multiple tasks). We measure deployment cost as the amount of
energy and carbon required to perform 1,000 inferences on repres
Shell cwd was reset to ~/Documents/College/U ... [truncated]
```

[Tool: Bash — Map index.html structure and IDs]

```
18:<article class="aipf" id="aipf">
44:  <nav class="aipf-nav" id="aipf-nav" aria-label="Sections">
47:    <span class="aipf-nav-now-fig" id="aipf-nav-total" aria-
live="polite">&mdash;</span>
48:    <span class="aipf-nav-now-label" id="aipf-nav-label">today</span>
66:    <button type="button" class="aipf-report-btn" id="aipf-report"
74:  <section class="aipf-panel" id="session" aria-labell ... [truncated]
```

[Tool: Bash — Read methodology start and tracker CSS]

```
<details class="aipf-method" id="methodology">
  <summary>How these numbers are made</summary>
  <div class="aipf-method-body">
    <p>
      <strong>Per-model AI estimates.</strong> The energy, carbon, and water numbers
for each
      prompt come from the open-source <a href="https://ecologits.ai/"
target="_blank" rel="noopener">EcoLogits</a>
      library (v0.10), the same code that ... [truncated]
```

[Tool: Bash — See how budget/team use the log toggle]

```
assets/js/nav.js:93:    const driving = typeof AIPF.trackerIsLogDriving === 'function'
&& AIPF.trackerIsLogDriving();
assets/js/nav.js:94:    box.textContent =
AIPF.fmtMetric(AIPF.aiDaily(AIPF.state.metric));
assets/js/budget.js:58: function monthToDate(metric) {
assets/js/budget.js:162:    const mtd = monthToDate(metric);
assets/js/team.js:66:    const you = AIPF.aiDaily(metric);
assets/js/team. ... [truncated]
```

[Tool: Edit — data.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/data.js has been updated successfully.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

[Tool: Edit — core.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/core.js has been updated successfully.

Now the tracker:

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

Now the HTML for the image form and its note:

[Tool: Read — index.html]

```
92
93     <div class="aipf-tk-form">
94         <p class="aipf-tk-form-h">Log a prompt</p>
95         <div class="aipf-tk-controls">
96             <select id="aipf-tk-model" aria-label="Model to log"></select>
97             <select id="aipf-tk-size" aria-label="Output length to log"></select>
98             <button type="button" class="aipf-tk-btn" id="aipf-tk-log">+
Log</button>
99 ... [truncated]
```

[Tool: Bash — Read tracker form CSS]

```
.aipf-tk-live { font-family: var(--font-body); font-size: 0.78rem; color: var(--text-
secondary); margin: 0.6rem 0 0; }
.aipf-tk-pace { font-family: var(--font-body); font-size: 0.78rem; color: var(--dim);
margin: 0.2rem 0 0; font-variant-numeric: tabular-nums; }

/* Log controls */
.aipf-tk-form { margin-top: 1rem; padding-top: 0.9rem; border-top: 1px solid var(--
border); }
.aipf-tk-form-h { font- ... [truncated]
```

[Tool: Edit — index.html]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/index.html has been updated successfully.

[Tool: Edit — features.css]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/css/features.css has been updated successfully.

[Tool: Bash — Check CSS tokens and methodology paragraph list]

```
--accent-hover: --accent-subtle: --accent: --bg-elevated: --bg-subtle: --bg: --border-
strong: --border: --dim: --font-body: --font-editorial: --max-width: --ok-subtle: --ok:
--on-accent: --text-secondary: --text: --warn-subtle: --warn:
assets/css/base.css:143:.aipf select {
assets/css/base.css:181:.aipf-life select {
assets/css/features.css:67:.aipf-tk-controls select { flex: 1 1 8rem; min-width: ...
[truncated]
```

[Tool: Read — index.html]

346 Output, words, and code. Each row maps to an EcoLogits output-token count:

```
347         tweet (50), short email (170), article summary (250), chatbot reply (400),
5-page report
348         (5,000), long document (15,000), coding or agent session (100,000), and re-
writing the
349         Lord of the Rings trilogy (500,000) – both EcoLogits benchmark tasks; the
trilogy it ... [truncated]
```

Adding a methodology paragraph for images after "Output, words, and code":

[Tool: Edit — index.html]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief – AI Footprint Calculator/index.html has been updated
successfully.
```

[Tool: Edit — index.html]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief – AI Footprint Calculator/index.html has been updated
successfully.
```

Now the calculator's running line and the cited report:

[Tool: Edit — calculator.js]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/calculator.js has been
updated successfully.
```

[Tool: Edit — calculator.js]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/calculator.js has been
updated successfully.
```

Now the report's header and an image note with citations:

[Tool: Bash — Locate report header and footnote line]

```
345:      '<table><thead><tr><th>Model</th><th>Typical output</th><th>
class="n">Prompts/day</th><th class="n">CO2e/day
347:      '<p class="muted">Per-prompt figures come from the EcoLogits model' + fn(1) +
'; the electricity is costed on
```

[Tool: Bash — Show the report footnote line]

```
'<p class="muted">Per-prompt figures come from the EcoLogits model' + fn(1) + '; the
electricity is costed on the ' + escXml(loc.label) + ' grid' + fn(2) + ', and
EcoLogits' embodied hardware emissions are kept.</p>' +
```

[Tool: Bash — Add image note and references to the report]

```
checked
```

Removing the unused IMAGE_SOURCES constant I added earlier, since the links live in the HTML:

[Tool: Bash — Remove unused image source constant]

```
['gpt-5.5', 'chat', 15],
  ['claude-sonnet-4-6', 'chat', 8],
  ['gemini-3.5-flash', 'summary', 4],
];

AIPF.WORLD_GRID = 480; // global-average grid carbon intensity, g CO2e/kWh (Ember)

// ---- Generated images (feature 1): one general range, Wh per image ----
// Central: median across Luccioni et al. 2024's image models (1.35 kWh per
// 1,000 inferences, 2022-23 open models on an A ... [truncated]
```

Running feature 1's acceptance checks in the preview:

[Tool: Browser navigate — http://localhost:4173/]

```
navigated to http://localhost:4173/
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "check1_oneImage": {
    "c": [
      0.513,
      0.0327,
      4.3662
    ],
    "e": [
      1.35,
      0.086,
      11.49
    ],
    "wML": [
      6.588,
      0.42,
      56.071
    ]
  },
  "check2_fourMore": {
```

```

    "c": [
      2.052,
      0.1307,
      17.4648
    ],
    "e": [
      5.4,
      0.344,
      45.96
    ],
    "wml": [
      26.352,
      1.679,
      224.285
    ]
  }, ... [truncated]

```

Checks 1–3 pass. Now check 4 (refresh, recent list, history, undo) and check 5 (the log toggle and the budget):

[Tool: Browser navigate — <http://localhost:4173/>]

navigated to <http://localhost:4173/>

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```

{
  "afterRefresh": {
    "count": "2 prompts · 5 images",
    "recent": [
      "Generated image",
      "Generated image",
      "Generated image"
    ]
  },
  "afterUndo": {
    "count": "2 prompts · 4 images",
    "dayTotal": 2.704036,
    "stored": 6
  },
  "budget0n": "3.22 g CO2e of 4.38 kg CO2e · day 26 of 30",
  "dayTotal0n": 3.217036,
  "historyTodayTitle": "2026-09-26: 3.22 g CO2e",
  " ... [truncated]

```

Checks 4 and 5 pass. Now check 6 (links and labels), the report, and the layout:

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "imageNote": [
    "Each generated image is about 1.35 Wh (range 0.086 Wh to 11.5 Wh), from open
models on research hardware, because no commercial image tool publishes figures[11]
[12]. Image carbon is that electricity on the same grid, with no embodied carbon; image
water is derived at 4.88 L/kWh, the US data-centre average[13].",
  ],
  "imageRow": [
    "Generated images (logged)One general ... [truncated]"
  ]
}
```

[Tool: Browser find — Log generated images]

Found 1 match(es) for "Log generated images":
– generic "Log generated images" [ref_66]

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — scroll_to]

scrolled ref_66 into view

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x609

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — scroll]

scrolled down at (400, 450)

Tab Context:

- Executed on tabId: seed

- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x600

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Working. Checking phone width and dark mode:

[Tool: Browser resize_window — mobile]

Viewport set to 375x812 (mobile) on this tab (scaled down to fit if larger than the pane). Reset it with preset "desktop" as soon as you finish testing; the desktop app also clears a size you set when your turn ends or the pane's width changes, so set it again if you still need it. If the user wants to keep a phone or tablet size after your turn, they can pick Mobile or Tablet in the pane's Viewpo ... [truncated]

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "iw": 375,
  "overflow": false,
  "sw": 375
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

– Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser resize_window — desktop]

Viewport emulation cleared; the tab is back to the pane's responsive size (desktop). Color scheme emulation set to light on this tab; it survives reloads until you set the other value or the pane re-syncs the tab to the app theme.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

"cleared"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Updating the plan's checkpoint 2 tasks:

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

Checkpoint 2 (image logging) is built, and all six of feature 1's acceptance checks passed in the preview. Before you verify it, I need one decision from you (below).

What changed - Data (data.js): the image figures (1.35 Wh per image, range 0.086–11.49 Wh) and the 4.88 L/kWh water factor, with comments naming their sources. I opened all five source links directly and found the figures in the source text. - **Costing (core.js):** image carbon is the image's energy on the selected region's grid, with no embodied carbon. Image water is the image's energy × 4.88 L/kWh. -

Tracker ([tracker.js](#), [index.html](#)): - A new "Log generated images" form takes a count, including discarded variations. - It saves one entry per image, so "Undo last" removes one image, the same way it removes one prompt. - The count line now reads, for example, "2 prompts · 4 images". - Below the form, energy, carbon and water are shown for one image and for today's images, each as a central value with its range. - A plain-language note links to S1, S2, S3 and S46. - **Elsewhere:** - The calculator's running line mentions images. - The report gets an image row, a short note, and 3 new references. - The methodology section has a new "Generated images" paragraph.

Acceptance checks (US region)

#	Check	Result
1	One image	1.35 Wh (0.086–11.49) · 0.513 g (0.033–4.37 g) · 6.59 mL (0.42–56.1 mL) ✓
2	N images	4 more images added exactly 4× those amounts ✓
3	Region change	UK and India change carbon by their grid intensity; energy and water unchanged ✓
4	Persistence	Images survive a refresh, show in the recent list and 7-day history, and "Undo last" removes one ✓
5	Log toggle	On: images are in the AI total and the budget. Off: excluded from both, and marked "not counted" ✓
6	Sources and labels	Every figure has a link, and the result shows its range and the limitations ✓

The layout also works at phone width and in dark mode, with no horizontal scroll. There are no console errors, and I cleared the test data afterwards.

Decision needed. Since Project 1, logged prompts count in the monthly budget even when "Use my log" is off. As the spec requires, images now don't. So with the toggle off, the budget counts your logged prompts but not your logged images. Options: - **A.** Leave it as built. The spec is met, but prompts and images behave differently in the budget. - **B.** Make logged prompts follow the toggle too. This changes Project 1's budget behaviour, so I'd add a revision note in the spec. - **C.** Count images in the budget regardless of the toggle, like prompts. This departs from acceptance check 5, so it would also need a revision note in the spec.

Which do you want? After that, verify checkpoint 2 and tell me when to commit it.

5. Learner (RG Sureshkumar)

go with b and then commit checkpoint 2

Agent

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Bash — Find related labels and plan revisions section]

```
assets/js/tracker.js:266:      line.appendChild(el('span', 'aipf-tk-entry-val',
counts(e) ? AIPF.fmtMetric(entryCost(e, AIPF.state.metric)) : 'not counted'));
assets/js/tracker.js:310:      (tracker.useLog ? '' : ' <em>Not counted while "Use my
log" is off.</em>');
145:## Revisions
146-
147-Record material changes to the approach, sequence, or checklist and explain why
they were made.
148-
```

Adding a short line to the tracker headline so a zero total isn't confusing when the toggle is off:

[Tool: Edit — tracker.js]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/tracker.js has been updated successfully.

[Tool: Edit — spec.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/spec.md has been updated successfully.

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

Verifying in the preview before committing:

[Tool: Browser navigate — <http://localhost:4173/>]

navigated to <http://localhost:4173/>

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "back": {
    "budget": "1.17 g CO2e of 4.38 kg CO2e · day 26 of 30",
    "tk": "1.17 g CO2e"
  },
  "off": {
    "ai": 20.847521999999998,
    "budget": "0 g CO2e of 4.38 kg CO2e · day 26 of 30",
    "day": 0,
    "live": "Last logged just now. "Use my log" is off, so nothing logged is counted.",
    "recent": "not counted,not counted,not counted",
    "tk": "0 g CO2e"
  },
  "on": {
    "ai" ... [truncated]
```

Works as intended. Committing:

[Tool: Bash — Commit checkpoint 2 on project-2]

```
M assets/css/features.css
M assets/js/calculator.js
M assets/js/core.js
M assets/js/data.js
M assets/js/tracker.js
M index.html
M plan.md
M spec.md
397e7eb Add image generation logging (feature 1)
```

I made logged prompts follow "Use my log" (option B) and committed checkpoint 2 on project-2 as 397e7eb. Nothing is pushed.

What option B changed - While "Use my log" is off, no logged entry of any kind counts: - The tracker total, the 7-day history and the monthly budget all read 0. - The recent list marks each entry "not counted". - When the toggle is off and entries exist, a new line under the tracker total reads: "“Use my log” is off, so nothing logged is counted." I added it so a zero total doesn't look like a bug. - With the toggle on, nothing changes: the tracker total and the budget both read 1.17 g for my test entries. - I checked all of this in the preview and cleared the test data afterwards.

Records: spec.md and plan.md each have a Revisions entry for this decision, and plan.md has a new, ticked checkpoint 2 task for it. The acceptance checks are unchanged.

"User verifies checkpoint 2" is still unticked. Tell me once you've checked it and I'll tick it.

Next is checkpoint 3 (agent sessions and projects), the largest step so far: - a session form with a model choice, four token fields and a project name - a breakdown of each session - a project view and a project-estimate form pre-filled with the S59 example - complete removal of the "agent" output size, including the team data and old share links - then feature 2's acceptance checks 1–7

Should I start checkpoint 3?

6. Learner (RG Sureshkumar)

start checkpoint 3

Agent

[Tool: Bash — Read source entries for feature 2]

```
### S7. Bai et al. (2026)
```

- Citation: Bai, L., Huang, Z., Wang, X., Sun, J., Mihalcea, R., Brynjolfsson, E., Pentland, A., & Pei, J. (2026). How Do AI Agents Spend Your Money? Analyzing and Predicting Token Consumption in Agentic Coding Tasks. arXiv:2604.22750. <<https://arxiv.org/abs/2604.22750>>
- Claim or figure: agentic coding consumes about 1,000× more tokens than code reasoning and code chat, ... [truncated]

Checking the feature 2 source links and the key figures:

[Tool: Bash — Check feature 2 source URLs and key figures]

```
200 https://arxiv.org/abs/2604.22750
200 https://simonpcouch.com/blog/2026-01-20-cc-impact/
200 https://ecologits.ai/latest/methodology/llm_inference/
200 https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use
200 https://arxiv.org/abs/2505.09598
200 https://arxiv.org/abs/2607.13080
200 https://arxiv.org/abs/2505.23970
200 https://arxiv.org/abs/2601.22076
== s8.txt 32717
592, ... [truncated]
```

[Tool: Bash — Find S10 input-cost figures and S8 rates]

```
would substantially increase the cost per query to around -> tokens (similar to a
short paper or long magazine article) would substantially increase the cost per query
to around 2.5 watt-hours , and a very long input length of 100k tokens (equal to
roughly 200 pages of text) would require almost 40 watt-hours of energy. Note that this

100k tokens -> to around 2.5 watt-hours , and a very long i ... [truncated]
```

[Tool: Bash — Find agent-size usages and share-link code]

```
assets/js/budget.js:19: // never across providers: moving an agent session from Opus
straight to
assets/js/calculator.js:196: const wd = AIPF.dailyWords(), cd =
AIPF.dailyCodeLines();
```

```
assets/js/core.js:15:      SIZE_LABEL_OVERRIDES, GAL_TO_L, DAYS,
assets/js/core.js:16:      TOKENS_PER_WORD, TOKENS_PER_LINE, CODE_FRACTION } =
AIPF;
assets/js/core.js:33:    const o = SIZE_LABEL_OVERRIDE ... [truncated]
```

[Tool: Bash — Read budget swaps, team data, nav sections, hint]

```
*/
(function (AIPF) {
  'use strict';

  const { el, store } = AIPF;

  // The next model down within the same provider's family. Only one step, and
  // never across providers: moving an agent session from Opus straight to
  // Haiku would show a far bigger saving, but it is not a substitution most
  // teams can actually make, and advice nobody can follow is not advice.
  // Someone who takes th ... [truncated]
```

[Tool: Bash — Read nav links and end of session panel]

```
<nav class="aipf-nav" id="aipf-nav" aria-label="Sections">
  <div class="aipf-nav-in">
    <div class="aipf-nav-now">
      <span class="aipf-nav-now-fig" id="aipf-nav-total" aria-live="polite">&mdash;
    </span>
      <span class="aipf-nav-now-label" id="aipf-nav-label">today</span>
    </div>
    <ul class="aipf-nav-links">
      <li><a class="aipf-nav-link" data-target="session" href= ... [truncated]
```

Starting with the data: the session rates, the S59 example, and removing the "agent" size.

[Tool: Bash — Read the sizes block in data.js]

```
// ---- Per-prompt model impacts from EcoLogits v0.10 (mean + 95% range) ----
// wh = electricity (Wh), emb = embodied hardware carbon (g CO2e), ml = water (mL)
AIPF.MODELS = [{"id":"gpt-5.5","name":"GPT-5.5","group":"OpenAI","sizes":{"tweet":
{"wh":0.4064,"whmin":0.2917,"whmax":0.5212,"emb":0.0241,"embmin":0.0241,"emb

AIPF.SIZES = [
  { id: 'tweet',    label: 'A tweet',                wor ... [truncated]
```

[Tool: Bash — Add session data, remove agent size and team rows]

```
assets/js/data.js:16:  // Each model's "agent" entry is EcoLogits' 100,000-output-token
"assist
assets/js/data.js:17:  // application development" benchmark. It is no longer an output
size; agent
assets/js/data.js:33:  // this older list, which still included the removed 'agent'
size.
```

```
assets/js/data.js:34: AIPF.LEGACY_SIZE_IDS = ['tweet', 'email', 'summary', 'chat',  
'report', 'long', 'agent', 'no ... [truncated]
```

Now `core.js`: session costing, guards against the removed size, and removal of the code-lines helpers, which only worked on "agent" rows.

[Tool: Bash — Read current core.js math section]

```
'use strict';  
  
const { MODELS, SIZES, LOCATIONS, HOMES, DRIVING, DIETS, FLYING,  
        SIZE_LABEL_OVERRIDES, GAL_TO_L, DAYS,  
        TOKENS_PER_WORD, TOKENS_PER_LINE, CODE_FRACTION } = AIPF;  
  
let uidSeq = 1;  
AIPF.nextUid = () => uidSeq++;  
  
// The single source of truth the whole app reads from.  
AIPF.state = {  
  rows: [], metric: 'carbon',  
  loc: 'us', home: 'med', drive: 'davg', ... [truncated]
```

[Tool: Bash — Add session math and agent-size guards to core]

```
ok
```

Checking the session maths in Node against the spec's acceptance values before building the UI:

[Tool: Bash — Compute acceptance values for sessions in Node]

```
check1 energy Wh [ 46.652, 17.686, 238.219 ] carbon g [ 18.178, 7.171, 90.973 ] water  
mL [ 201.72, 60.76, 1136.169 ]  
breakdown out [ 22.952, 15.686, 30.219 ] input [ 4.2, 2, 8 ] cacheRead [ 19.5, 0, 200 ]  
check5 kWh [ 708.306, 74.138, 6246.247 ] kg [ 270.972, 29.988, 2375.389 ] L [ 3351.95,  
258.81, 30375.505 ]  
sizes tweet,email,summary,chat,report,long,novel
```

[Tool: Bash — Update share links, words, running line, report]

```
assets/css/base.css:240:.aipf-words-code { display: block; margin-top: 0.7rem; }
```

[Tool: Bash — Remove the now-unused code-lines CSS rule]

```
0
```

Now the tracker:

[Tool: Bash — Update tracker for sessions and removed size]

```
ok
```

Now the new sessions.js for the session form, the project view and the project estimate:

[Tool: Write — sessions.js]

```
File created successfully at: ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint
Calculator/assets/js/sessions.js (file state is current in your context – no need to
Read it back)
```

Now the HTML: the session form in the tracker, a new "Agent projects" panel, the nav link, and the script tag.

[Tool: Bash — Add session form, projects panel, nav, script]

```
19:   for (const name of ['initTracker', 'initSessions', 'initTeam', 'initBudget',
'initNav']) {
```

Rewording the Couch note so it doesn't assume the author's pronouns:

[Tool: Bash — Replace gendered pronouns in the Couch note]

```
(Bash completed with no output)
```

Now the CSS. First the existing panel and grid patterns:

[Tool: Bash — Read panel, grid, and select CSS patterns]

```
/**
 * features.css – styles for the company features added on top of the original
 * calculator: the live session tracker, the team dashboard, and the budget.
 *
 * These reuse the tokens in base.css so the additions read as part of the same
 * page rather than bolted on.
 */

/* ---- Shared panel shell ---- */
.aipf-panel {
  background: var(--bg-elevated);
  border: 1px solid var(--border);
  p ... [truncated]
```

[Tool: Bash — Add CSS for session form and projects panel]

```
done
```

Now the methodology text: removing the old agent-size and code-lines wording, and adding an agent-sessions paragraph.

[Tool: Bash — Rewrite methodology for sessions; fix budget comment]

```
index.html:133:      <summary class="aipf-tk-form-h">Log an agent session</summary>
index.html:259:      <p class="aipf-hint">"Typical output" is how long each reply is.
Coding-agent sessions and generated images are logged
index.html:442:      An earlier version offered a "coding / agent session" row of
100,000 output tokens. It
index.html:443:      was removed because real agent session ... [truncated]
```

Now testing in the preview:

[Tool: Browser navigate — http://localhost:4173/]

navigated to http://localhost:4173/

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "check1": {
    "Wh": [
      46.65,
      17.69,
      238.22
    ],
    "g": [
      18.18,
      7.17,
      90.97
    ],
    "mL": [
      201.72,
      60.76,
      1136.17
    ]
  },
  "count": "1 prompt · 1 session",
  "last": "Last session logged · Website redesign · Claude Opus 4.8\n\n46.7 Wh (17.7
```

```
Wh-238 Wh) · 18.2 g CO2e (7.17 g CO2e-91.0 g CO2e) · 202 mL (60.8 mL-1.14 L)\n\nOutput\  
... [truncated]
```

Check 1 matches the spec. The running line reads "1 prompts, 1 agent session", so I'll tidy its wording:

[Tool: Edit — calculator.js]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI  
Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/calculator.js has been  
updated successfully. (note: the file had been modified on disk since you last read it  
– the edit applied cleanly, but the file contains other changes not in your context.  
Read it before edits that depend on surr ... [truncated]
```

Now checks 2, 4 and 5:

[Tool: Browser navigate — http://localhost:4173/]

```
navigated to http://localhost:4173/
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{  
  "check2": {  
    "base": {  
      "cr": 19.5,  
      "inp": 4.2,  
      "out": 22.952  
    },  
    "moreFresh": {  
      "cr": 19.5,  
      "inp": 6.3,  
      "out": 22.952  
    },  
    "moreOut": {  
      "cr": 19.5,  
      "inp": 4.2,  
      "out": 45.904  
    },  
    "moreRead": {  
      "cr": 39,  
      "inp": 4.2,  
      "out": 22.952  
    },  
    "moreWrite": {  
      "cr": 19.5,
```

```
"inp": 6.3,  
"out" ... [truncated]
```

Checks 2, 4 and 5 pass. Now check 3 (the removed size, old share links and team data) and check 6 (refresh, undo, budget, toggle):

[Tool: Browser javascript_tool — javascript_exec]

```
"seeded"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1")

[Tool: Browser navigate — http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1]

```
navigated to http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: TypeError: Cannot read properties of undefined (reading 'k')  
    at entryKind (http://localhost:4173/assets/js/core.js:135:30)  
    at Object.entryTriple (http://localhost:4173/assets/js/core.js:192:18)  
    at <anonymous>:4:29
```

[Tool: Browser javascript_tool — javascript_exec]

```
{  
  "mem": [  
    "prompt",  
    "session",  
    "session",  
    "session"  
  ],  
  "nav": 0,  
  "now": 26992.100000023842,  
  "stored": [  
    "prompt",  
    "agent",  
  ]  
}
```

```
"session",  
"session",  
"session"  
]  
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2. ... [truncated])

Right: a hash-only navigation didn't reload the page. Reloading properly:

[Tool: Browser javascript_tool — javascript_exec]

```
"reloading"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: ReferenceError: AIPF is not defined  
    at <anonymous>:1:12
```

Ran before the page loaded; retrying:

[Tool: Browser javascript_tool — javascript_exec]

```
{  
  "check6": {  
    "afterUndo": {  
      "sessions": 2,  
      "stored": 2  
    },  
    "budget": "25.2 g CO2e of 532 g CO2e · day 26 of 30",  
    "off": {  
      "budget": "0 g CO2e of 532 g CO2e · day 26 of 30",  
      "counted": 0,  
      "dayTotal": 0  
    },  
    "onAgain": {  
      "budget": "25.1 g CO2e of 532 g CO2e · day 26 of 30",  
      "counted": 2  
    }  
  },  
}
```

```
"sessionsAfterReload": 3,  
"tracke ... [truncated]
```

Checks 3 and 6 pass on the log side. Now old and new share links, with the log off so the URL's rows show:

[Tool: Browser javascript_tool — javascript_exec]

```
"ok"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1")

[Tool: Browser javascript_tool — javascript_exec]

```
{  
  "newLink": "k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=4-3-5.3-6-1&sv=2",  
  "oldLinkRows": [  
    "claude-sonnet-4-6:chat:5",  
    "claude-opus-4-8:novel:1"  
  ]  
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=3-6-2.4-3-5.3-7-1")

The old link skips its "agent" row and loads the trilogy row correctly. Round-tripping the new link:

[Tool: Browser javascript_tool — javascript_exec]

```
"ok"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition"
("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=4-3-5.3-6-1&sv=2")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: ReferenceError: AIPF is not defined  
    at <anonymous>:1:11
```

[Tool: Browser javascript_tool — javascript_exec]

```
[
  "claude-sonnet-4-6:chat:5",
  "claude-opus-4-8:novel:1"
]
```

Tab Context:

– Executed on tabId: seed

– Available tabs:

- tabId seed: "AI Footprint Calculator – Company Edition"

("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=4-3-5.3-6-1&sv=2")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

– Executed on tabId: seed

– Available tabs:

- tabId seed: "AI Footprint Calculator – Company Edition"

("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=4-3-5.3-6-1&sv=2")

New links round-trip correctly. Now check 7 (links) and a regression check on feature 1:

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "cacheLabel": true,
  "img": [
    0.513,
    0.0327,
    4.3662
  ],
  "methodology": [
    "https://arxiv.org/abs/2604.22750",
    "https://ecologits.ai/latest/methodology/llm_inference/",
    "https://epoch.ai/gradient-updates/how-much-energy-does-chatgpt-use",
    "https://arxiv.org/abs/2505.09598",
    "https://simonpcouch.com/blog/2026-01-20-cc-impact/",
    "https://eta-publications.lbl ... [truncated]"
  ]
}
```

All seven sources are linked and each returned 200 earlier. Image figures are unchanged. Now the visual check:

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/#k=carbon&l=us&h=med&v=davg&d=avg&f=some&r=4-3-5.3-6-1&sv=2")

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x609

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — scroll]

scrolled down at (400, 300)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x600

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

The breakdown rows spread across three columns, with the central value floating in the middle. I'll keep each value next to its range:

[Tool: Bash — Group breakdown value with its range]

ok

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: TypeError: Cannot read properties of null (reading  
'scrollIntoView')  
    at <anonymous>:1:37
```

[Tool: Browser computer — scroll]

scrolled down at (400, 300)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x600

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

The projects panel looks right. Checking phone width in dark mode:

[Tool: Browser resize_window — mobile]

Viewport set to 375x812 (mobile) on this tab (scaled down to fit if larger than the pane). Reset it with preset "desktop" as soon as you finish testing; the desktop app also clears a size you set when your turn ends or the pane's width changes, so set it again if you still need it. If the user wants to keep a phone or tablet size after your turn, they can pick Mobile or Tablet in the pane's Viewpo ... [truncated]

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "iw": 375,
  "overflow": false,
  "sw": 375
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — scroll]

scroll with `coordinate` requires a prior computer{action:"screenshot"} (no screenshot dimensions cached)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "overflow": false
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

Works at phone width in dark mode. Resetting the viewport, re-running the feature 1 checks as a regression test, and clearing test data:

[Tool: Browser resize_window — desktop]

Viewport emulation cleared; the tab is back to the pane's responsive size (desktop). Color scheme emulation set to light on this tab; it survives reloads until you set the other value or the pane re-syncs the tab to the app theme.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

javascript_tool failed: ReferenceError: AIPF is not defined
at <anonymous>:1:12

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "afterUndo": 2,
  "off": 0,
  "perImage": {
    "L": [
      0.0066,
      0.0004,
      0.0561
    ],
    "Wh": [
      1.35,
      0.086,
      11.49
    ],
    "g": [
      0.513,
      0.0327,
      4.3662
    ]
  }
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Feature 1 still passes. Reloading so the cleared state takes effect, then updating the plan:

[Tool: Browser javascript_tool — javascript_exec]

```
"ok"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Edit — plan.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/plan.md has been updated successfully.

[Tool: Bash — Syntax-check scripts and summarize changes]

```
M assets/css/base.css
M assets/css/features.css
```

```

M assets/js/app.js
M assets/js/budget.js
M assets/js/calculator.js
M assets/js/core.js
M assets/js/data.js
M assets/js/team-data.js
M assets/js/tracker.js
M index.html
M plan.md
?? assets/js/sessions.js
assets/css/base.css      | 1 -
assets/css/features.css | 34 ++++++++--
assets/js/app.js         | 2 +-
assets/js/budget.js      | ... [truncated]

```

Checkpoint 3 (agent sessions and projects) is built, and all seven of feature 2's acceptance checks passed in the preview. Nothing is committed yet, and two side effects of removing the "agent" size need your attention first.

What changed - Session maths ([core.js](#), [data.js](#)): - Output tokens use the model's EcoLogits "agent" figure ÷ 100,000. - Fresh input and cache writes are costed at 2.1 Wh per 10,000 tokens (range 1–4). - Cache reads are costed at 0.39 Wh per 10,000 (range 0–4). - Carbon includes embodied carbon for the output tokens only. Water is EcoLogits' own figure for the output, plus 4.88 L/kWh for the input side. - I opened all 8 source links and found the figures in the text: S8's 592,439 tokens, 24 calls, 41 Wh and 39 Wh/MTok; S10's 2.4 Wh against 0.3 Wh; S59's four token counts; S7's "up to 30x". - **Session form** (tracker): the "Agent session..." button opens a form with a model, the four token fields (empty) and an optional project name. - After logging, it shows the session's energy, carbon and water with ranges, plus a breakdown: output, fresh input and cache writes, and cache reads labelled "our estimate". - The form includes the S8 reference note and the limitations, with links to S7–S11 and S46. - **"Agent projects" panel** (new [sessions.js](#), plus a "Projects" nav link): - The project view adds up each project's sessions across the last 120 days. - The estimate form is pre-filled with the labelled S59 example. Its result is shown beside the AI total, with a statement that it's not added to it. - **Removed:** the "agent" size in the calculator, the tracker and the team data. Old log entries of that size show "removed size · not counted". Old share links skip that row and read the others correctly; new links carry a version marker (sv=2). - **Methodology:** I rewrote the "Output and words" paragraph and added an "Agent sessions and projects" paragraph. The report gets session rows and 3 new references.

Acceptance checks (US region, Claude Opus 4.8)

#	Result
1	46.65 Wh (17.69–238.2) · 18.18 g (7.17–90.97) · 201.7 mL (60.8–1,136); breakdown 22.95 / 4.2 / 19.5 Wh ✓
2	Raising one token type raises only its own part ✓
3	No "agent" size in any menu or the team data; old entries count nothing; an old link loads "Sonnet chat ×5, Opus trilogy ×1" and skips the agent row ✓
4	3 sessions tagged across 2 days are listed together, and the total equals their sum; each counts once in its own day ✓

#	Result
5	S59 pre-fill gives 708 kWh (74–6,246) · 271 kg (30–2,375) · 3,352 L (259–30,376); editing it doesn't change the AI total ✓
6	Sessions survive a refresh, undo removes one, and they're in the budget when the log is on and out when it's off ✓
7	Every figure links to its source; cache reads are labelled "our own estimate; no measurement exists" ✓

I also re-ran feature 1's checks (they still pass) and checked the layout at phone width and in dark mode. There are no console errors, and I cleared the test data.

Side effects to review: 1. **The "lines of code" estimate is gone.** It worked only on "agent" rows, so the "What it writes" box now shows words only. 2. **The default monthly budget fell sharply, from 4.38 kg to 532 g CO₂e.** That default comes from the team sample averages, and the "agent" rows you decided to delete made up most of them. This follows from your decision in the spec, but the new default may look small. It applies only until someone sets their own budget.

I ticked the agent tasks for checkpoint 3 in `plan.md`; "User verifies checkpoint 3" is left for you. Are both side effects OK, and should I commit checkpoint 3?

7. Learner (RG Sureshkumar)

the side effects are ok and you commit checkpoint 3 and then start checkpoint 4

Agent

[Tool: Bash — Commit checkpoint 3 on project-2]

```
3210da9 Add agent sessions and projects (feature 2)
```

Committed checkpoint 3 as 3210da9. Starting checkpoint 4 (the digital day builder) by reading its sources:

[Tool: Bash — Read source entries for feature 3]

```
### S19. Mytton, Lundén & Malmodin (2024)
```

```
- Citation: Mytton, D., Lundén, D., & Malmodin, J. (2024). Network energy use not
directly proportional to data volume: The power model approach for more reliable
network energy consumption calculations. *Journal of Industrial Ecology*, 28(4), 966–
980. <https://doi.org/10.1111/jiec.13512>
- Claim or figure: network energy consumption is not proportional t ... [truncated]
```

The feature 3 source entries are read. Before writing code, I'll check the S25 white paper's figures directly, since most of the device, network and data values come from it:

[Tool: Bash — Download S25 and check its key figures]

```
200 5277639
```

```
fontTools is required to fully parse the encoding of a CFF Type1 font in font
dictionary {'/BaseFont': '/CEJK0J+MyriadPro-Regular', '/Encoding': '/WinAnsiEncoding',
'/FirstChar': 67, '/FontDescriptor': IndirectObject(712, 0, 4364189648), '/LastChar':
82, '/Subtype': '/Type1', '/Type': '/Font', '/Widths': [580, 0, 0, 0, 0, 0, 0, 0, 0,
0, 658, 689, 532, 0, 538]}, but is not installed. ... [truncated]
```

Most figures match. The laptop (22 W) and desktop (115 W) figures didn't match my search pattern, so I'm looking for them directly:

[Tool: Bash — Search S25 for device power and data-centre figures]

```
data cent\w+.{0,60}Wh -> dary for calculations. Masanet et al. (2020a) estimates that
since 2010, global data centre energy use has only increased by 6% since 2010 to 205TWh
in 2018. Where, this small rise in energy use occurred desp
```

```
data cent\w+.{0,60}Wh -> ining, in its reporting. Malmodin (2020) estimated the total
electricity use of data centres at 208TWh in 2018, or 0.9% of final global elec ...
[truncated]
```

[Tool: Bash — Print S25 device power table context]

```
information available from DIMPACT members. Assumed 40% of TV viewing occurs on smart
TVs which require no peripheral, 55% on TVs via set-top box and the remaining 5% on TVs
via gaming console. • Non-TV viewing assumed to be 6% smartphones, 12% desktops and 12%
laptops based on Comscore's The US Total Video Report (Comscore, 2014). • Android and
iPhone share of smartphones estimated based on Statc ... [truncated]
```

Every S25 device figure is confirmed in its Table 10: phone 1 W, desktop with monitor 115 W, laptop 22 W, TV 100 W. Now the other nine links and their figures:

[Tool: Bash — Check remaining feature 3 links and figures]

```
403 https://doi.org/10.1111/jiec.13512
200 https://www.cablelabs.com/blog/hourly-data-consumption-of-popular-video-
conferencing-applications
200 https://sustainability.atmeta.com/asset/2025-sustainability-report/
200 https://sustainability.atmeta.com/asset/2025-environmental-data-index/
200 https://www.playstation.com/en-gb/legal/ecodesign/
200 https://arxiv.org/abs/2402.06346
200 https://a ... [truncated]
```

Every figure matches. S19's DOI returns 403 because Wiley blocks automated requests; research.md already noted this. I'll confirm the DOI resolves through Crossref:

[Tool: Bash — Confirm the S19 DOI metadata via Crossref]

```
Network energy use not directly proportional to data volume: The power model approach
for more reliable network energy consumption calculations | Journal of Industrial
Ecology 28 4 966–980 | https://doi.org/10.1111/jiec.13512 |
```

<https://onlinelibrary.wiley.com/doi/10.1111/jiec.13512>
302

The DOI redirects correctly (302 to Wiley). Last, S32's 0.0055 MWh per daily active person:

[Tool: Bash — Verify Meta electricity intensity figure]

```
200 application/pdf 2266742
Shell cwd was reset to ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator
```

[Tool: Bash — Search Meta data index for per-person intensity]

```
35709
-- 160 3,154 Environmental Footprint 1,2,3,4,5 GHG Intensity Market-Based Scope 1 and
2 Emissions (in metric tons CO2 e/unit of key performance indicators) 2020 2021 2022
2023 2024 GHG intensity per daily active person 0.00002 0.00002 0.00002 0.000016 0.0000
--
(WUE) 2020 2021 2022 2023 2024 Annual data center WUE 0.30 0.26 0.20 0.18 0.19 Water
Withdrawal Intensity (in liters/unit of key p ... [truncated]
```

[Tool: Bash — Find Meta electricity intensity row]

```
MWh) 2020 2021 2022 2023 2024 2025 Environmental Data Inde x G 2.1 Electricity
Electricity Intensity (in MWh/ unit of key performance indicators) 2020 2 021 2 022 2
023 2 024 Electricity intensity per daily activ e person - 0 .0033 0 .0039 0 .0036 0
.0055 Electricity intensity per million USD revenue - 7 9.9 9 8.7 131.4 2 112. 00
Electricity Mix Matching (in % of total electricity consumed) 2020 2 ... [truncated]
```

All the feature 3 figures check out, including S32's 0.0055 MWh per daily active person for 2024 (about 15 Wh a day). Three points in the spec are open, and I'd rather you decide them than me guess:

[Tool: AskUserQuestion — Console W / Router mix / Log window]

```
Your questions have been answered: "The spec gives the console's streaming/menus draw
only as a range, 43–47 W (S33: 43.4 W and 47.06 W on the PS5 home menu), with no
central value. Which central value should the calculator use?"="Midpoint, about 45 W
(Recommended)", "When today's log replaces some activities and the typical day supplies
others, how should router hours be counted?"="Add the two parts (Recommended)", "A
logged digital activity has a duration but no start time. What time window should it
cover for the router–overlap calculation?"="Ends when logged (Recommended)". You can
now continue with these answers in mind.
```

[Tool: Edit — data.js]

```
The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI
Explorations/Unit 1 Brief – AI Footprint Calculator/assets/js/data.js has been updated
successfully. (note: the file had been modified on disk since you last read it – the
edit applied cleanly, but the file contains other changes not in your context. Read it
before edits that depend on surroundin ... [truncated]
```

Now the feature module, `digital.js`:

[Tool: Write — digital.js]

```
File created successfully at: ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint
Calculator/assets/js/digital.js (file state is current in your context – no need to
Read it back)
```

Now the tracker's handling of logged digital activities:

[Tool: Bash — Show digital activities in the tracker]

```
ok
```

Now the HTML: the tracker's log form, the "Your digital day" panel, the nav link, the script tag, and the init call.

[Tool: Bash — Add digital-day panel, log form, nav, script]

```
ok
```

Rewording one clumsy sentence in that note:

[Tool: Bash — Reword network sentence; find tag CSS]

```
assets/css/base.css:251:.aipf-tag { font-family: var(--font-body); font-size: 0.62rem;
font-weight: 600; text-transform: uppercase; letter-spacing: 0.04em; white-space:
nowrap; }
assets/css/base.css:252:.aipf-tag--save { color: var(--accent); }
```

Now the CSS for the digital-day panel:

[Tool: Bash — Add CSS for digital-day panel]

```
ok
```

Adding the methodology paragraph:

[Tool: Bash — Add digital-day methodology paragraph]

```
ok
```

Now testing. First, feature 3's acceptance checks 1–3 with the spec's typical day:

[Tool: Browser navigate — http://localhost:4173/]

```
navigated to http://localhost:4173/
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
{
  "check1": {
    "L": [
      1.603,
      1.521,
      1.685
    ],
    "Wh": [
      366.42,
      347.57,
      385.27
    ],
    "g": [
      139.24,
      132.08,
      146.4
    ]
  },
  "check2": {
    "byActivity": {
      "call": {
        "dc": [
          0,
          0,
          0
        ],
        "device": [
          44,
          44,
          44
        ]
      },
      "network": ... [truncated]
    }
  }
}
```

Checks 1–3 pass: 366.4 Wh (347.6–385.3), 139.2 g (132.1–146.4), 1.60 L (1.52–1.69), and the breakdown matches. Now checks 4–7:

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "check4": {
    "acts": [
      "call:typical:2h",
      "social:typical:1h",
      "stream:log:0.5h"
    ],
    "mixNote": true,
    "msg": "Logged. Today's video streaming now come from your log.",
    "recent": "8:13 PM | Video streaming · TV · 30 min | digital day",
    "router": [
      25,
      25,
      25
    ],
    "routerHours": 2.5,
    "tags": [
      "Video calls TYPICAL DAY 69.3 ... [truncated]"
    ]
  }
}
```

[Tool: Bash — Show isLogDriving definition]

```
114- // True when the log is driving the page and actually has something in it.
115- // With the log on but empty – a first visit – the page would otherwise be a
116- // wall of zeros, which reads as broken rather than as "nothing yet", so fall
117- // back to the original's example day until there is real usage to show.
118: function isLogDriving() { return tracker.useLog && tracker.today.le ... [truncated]
```

[Tool: Bash — Base log-driving mode on AI entries only]

```
158:   if (!tracker.today.length) return;
171:   if (tracker.today.length < 2) return null;
210:   const n = tracker.today.length;
225:   live.textContent = 'Last logged ' + agoText(tracker.today[n - 1].t) + '.' +
261:   if (!tracker.today.length) {
292:     if (tracker.today.length > RECENT_SHOWN) {
293:       box.appendChild(el('p', 'aipf-tk-more', '+' + (tracker.today.length -
RECENT_SHOWN) ... [truncated]
```

Re-running check 7 from a clean log:

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: ReferenceError: AIPF is not defined
  at <anonymous>:1:12
```

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "count": "2 prompts · 2 digital activities",
  "digitalTotal": [
    210.84,
    191.99,
    229.69
  ],
  "exampleDay": {
    "afterDigitalLog": true,
    "afterTypicalEdit": true,
    "driving": false
  },
  "loggedDay": {
    "afterDigitalLog": true
  },
  "msg": "Logged. Today's hours of video streaming now come from your log."
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs: ... [truncated]

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "cards": [
    "YOUR AI TOTAL TODAY | 1.56 Wh (1.17 Wh–1.95 Wh) | 0.652 g CO2e (0.503 g CO2e–0.801 g CO2e) | 5.82 mL (3.80 mL–7.84 mL) | Prompts, images, and agent sessions. Counts data centres only, not your own device or router.",
    "YOUR DIGITAL DAY | 211 Wh (192 Wh–230 Wh) | 80.1 g CO2e (73.0 g CO2e–87.3 g CO2e) | 926 mL (844 mL–1.01 L) | Your devices, home router, network, and data cent ... [truncated]"
  ]
}
```

All 10 required sources are linked. The phone, laptop, desktop and TV lines carry "single estimate, no range available". Checking the layout:

[Tool: Browser javascript_tool — javascript_exec]

```
"ok"
```

Tab Context:

- Executed on tabId: seed

- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer – screenshot]

[screenshot]

Screenshot size: 800x609

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer – scroll]

scrolled down at (400, 300)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer – screenshot]

[screenshot]

Screenshot size: 800x600

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

The desktop layout works. Checking phone width in dark mode:

[Tool: Browser resize_window – mobile]

Viewport set to 375x812 (mobile) on this tab (scaled down to fit if larger than the pane). Reset it with preset "desktop" as soon as you finish testing; the desktop app also clears a size you set when your turn ends or the pane's width changes, so set it again if you still need it. If the user wants to keep a phone or tablet size after your turn, they can pick Mobile or Tablet in the pane's Viewpo ... [truncated]

[Tool: Browser javascript_tool – javascript_exec]

```
{
  "overflow": false,
  "sw": 375
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — scroll]

scrolled down at (375, 800)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

The phone layout works too. Resetting the viewport, then running a regression pass on features 1 and 2 with an empty log:

[Tool: Browser resize_window — desktop]

Viewport emulation cleared; the tab is back to the pane's responsive size (desktop). Color scheme emulation set to light on this tab; it survives reloads until you set the other value or the pane re-syncs the tab to the app theme.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
javascript_tool failed: ReferenceError: AIPF is not defined
    at <anonymous>:1:12
```

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool – javascript_exec]

```
{
  "digitalEmpty": [
    0,
    0,
    0
  ],
  "emptyMsg": "Add an activity to your typical day, or log one in the tracker, to see your digital day.",
  "image": [
    [
      1.35,
      0.086,
      11.49
    ],
  ],
}
```

```
[
  0.513,
  0.0327,
  4.3662
],
[
  0.0066,
  0.0004,
  0.0561
]
],
"session": [
  [
    46.652,
    17.686,
    238.219
  ], ... [truncated]
```

Regression passes: image and session figures are unchanged. Recording your three decisions in the spec and updating the plan:

[Tool: Bash — Record decisions in spec and update plan]

```
M assets/css/features.css
M assets/js/app.js
M assets/js/data.js
M assets/js/tracker.js
M index.html
M plan.md
M spec.md
?? assets/js/digital.js
```

Checkpoint 3 is committed as 3210da9, and checkpoint 4 (the digital day builder) is built. All eight of feature 3's acceptance checks pass. Checkpoint 4 isn't committed yet.

Sources: I opened all 11 links and found the figures in the source text. The Carbon Trust's Table 10 gives phone 1 W, laptop 22 W, desktop with monitor 115 W and TV 100 W. Its network, router and data-use figures also match, as do Sony's PS5 wattages, 358.6 W (S36), 305.1 W (S37), Spotify's bitrates, CableLabs' 0.5–3.4 GB and Meta's 0.0055 MWh. S19's DOI blocks automated checks (403), but Crossref confirms it and it redirects correctly.

What changed - Data ([data.js](#)): device power, data use per hour, network intensities, router draw, data-centre figures and the 4.35 L/kWh water factor, with sources in the comments. - **"Your digital day" panel** (new [digital.js](#), plus a nav link): - A typical-day editor offers only the allowed combinations: cellular for the phone only, and quality where it applies. - The home Wi-Fi hours field defaults to the capped sum and can't go above it. - Your AI total and your digital day appear as two separate cards. There's a breakdown by component (devices, router, network, data centres) and by activity, with each activity tagged "typical day" or "today's log". - Phone, laptop, desktop and TV lines carry "single estimate, no range available". - The limitations note links all 10 required sources, and there's a methodology paragraph. - **Tracker:** a "Log a digital activity" form, and recent entries show, for example, "Video streaming · TV · 30 min · digital day". - **Your three decisions** are recorded in `spec.md`: a 45.2 W central

value for console streaming, router hours as the typical part plus the logged part, and logged activities treated as ending when logged.

Acceptance checks (US region, the spec's typical day)

#	Result
1	366.4 Wh (347.6–385.3) · 139.2 g (132.1–146.4) · 1.60 L (1.52–1.69) ✓
2	Laptop 44, TV 200, phone 1, router 40, network 64.35 (calls 25.35, streaming 39), data centres 17.07 (2 + 15.07) Wh ✓
3	Wi-Fi 4 → 3 h lowers the total by 10 Wh; entering 10 snaps back to the maximum of 4 ✓
4	A 30-minute HD TV stream replaces the typical streaming entry: total 170.7 Wh, router 2.5 h, origins labelled ✓
5	Social media set to 0 h removes its 15.07 Wh data-centre share, plus the phone's 1 Wh ✓
6	UK region: carbon × 125/380 exactly; energy and water unchanged ✓
7	AI total, budget, team view and nav total unchanged ✓ (after the fix below)
8	All 10 sources linked, and single-estimate labels present ✓

Bug found and fixed: check 7 failed at first. If a digital activity was the only thing logged today, the page switched to "log is driving" mode, so the AI total dropped from the example day to 0. Now only AI entries switch that mode. I also found that two overlapping logged streams count the router once.

I also re-ran features 1 and 2 (unchanged) and checked phone width and dark mode, with no horizontal scroll and no console errors. I cleared the test data afterwards.

Two things to check when you review: - As with option B, logged digital activities count only while "Use my log" is on. I applied this for consistency; it's noted in the spec revision. - The typical day is saved in your browser, so it persists between visits.

"User verifies checkpoint 4" is still unticked in `plan.md`. Should I commit checkpoint 4 and start checkpoint 5 (device manufacturing)?

8. Learner (RG Sureshkumar)

commit checkpoint 4 and start checkpoint 5

Agent

You've hit your session limit · resets 10:40pm (America/Chicago)

9. Learner (RG Sureshkumar)

commit checkpoint 4 and start checkpoint 5

Agent

[Tool: Bash — Commit checkpoint 4; read feature 4 sources]

```
c6624c1 Add the digital day builder (feature 3)
### S39. Apple, iPhone 17 Product Environmental Report (2025)

- Citation: Apple. (2025, September). *iPhone 17 Product Environmental Report*.
<https://www.apple.com/environment/pdf/products/iphone/iPhone_17_PER_Sept2025.pdf>
- Claim or figure: 55 kgCO2e (256GB) and 61 kgCO2e (512GB); breakdown for 256GB:
production materials and processes 53%, produ ... [truncated]
```

Committed checkpoint 4 as c6624c1. Starting checkpoint 5 by checking its 12 source links and the manufacturing figures in their text:

[Tool: Bash — Check feature 4 source links and figures]

```
200 s39
https://www.apple.com/environment/pdf/products/iphone/iPhone_17_PER_Sept2025.pdf
200 s40
https://www.apple.com/environment/pdf/products/notebooks/M4_MacBook_Air_PER_March2025.p
df
200 s41 https://devera.ai/benchmarks/carbon-footprint-of-a-laptop
200 s42
https://www.samsung.com/global/sustainability/media/pdf/TV_QLED_Environmental_Report_EN
.pdf
200 s43 https://www.samsung.com/global/sustaina ... [truncated]
```

[Tool: Bash — Look for the PS4 figure in S66]

```
\b89\b -> draws on results from Claire and John: "A few days later, Barlow
emailed me the results of her and Durrell's findings. The equivalent of 89
kilograms of carbon dioxide is emitted into the atmosphere with the production

carbon dioxide -> s from Claire and John: "A few days later, Barlow emailed me
the results of her and Durrell's findings. The equivalent of 89 k ... [truncated]
```

S66's 89 kg is confirmed; it's written as "89 kilograms". That leaves the S51 contradiction, which affects feature 4's acceptance check 4:

[Tool: AskUserQuestion — Phone water]

```
Your questions have been answered: "S51 states a smartphone's production water as
"3,190 gallons (12,760 liters)", but 3,190 gallons is about 12,075 L (and 12,760 L is
about 3,371 gallons). The approved spec uses 12,075 L (about 11.0 L a day over 3
years). Which figure should the calculator use?"="12,075 L, disclose the conflict
(Recommended)". You can now continue with these answers in mind.
```

You chose 12,075 L, with the conflict disclosed. Next, the methodology text for the existing yearly device comparisons:

[Tool: Bash — Find existing device comparison text]

```
281:      <a
href="https://www.carbontrust.com/sites/default/files/documents/resource/public/Carbon-
impact-of-video-streaming.pdf" target="_blank" rel="noopener">Carbon Trust (2021)</a>.
Your device dominates every activ
563:      (European averages for 2020, seed-funded by Netflix through DIMPACT).
Device power: smartphone
564:      1&nbsp;W, laptop 22&nbsp;W, office desktop with monitor 115& ... [truncated]
```

[Tool: Bash — Read the yearly comparison methodology text]

```
bulbs) and
      <a href="https://iopscience.iop.org/article/10.1088/1748-9326/ab8589"
target="_blank" rel="noopener">Ivanova et al. (2020)</a>
      (green electricity, heat-pump heating, a home retrofit, electric and hybrid
cars).
    </p>
    <p>
      <strong>Ways you add emissions (the yearly chart).</strong>
      Manufacturing a new car ≈ 6 t CO<sub>2</sub>e
      (<a href="ht ... [truncated]
```

Now the data: the eight device figures, plus updating the three yearly comparison items.

[Tool: Bash — Add device manufacturing data; update comparisons]

```
ok
```

Now devices.js:

[Tool: Write — devices.js]

```
File created successfully at: ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint
Calculator/assets/js/devices.js (file state is current in your context – no need to
Read it back)
```

Making the digital-day card show notes supplied by extra components, rather than hard-coding the manufacturing check:

[Tool: Bash — Generic card notes; register devices init]

```
ok
```

Now the HTML: the "My devices" block, its limitations note, the script tag, and the methodology text.

[Tool: Bash — Add My devices block, notes, methodology]

```
(Bash completed with no output)
```

Now the CSS:

[Tool: Bash — Add CSS for My devices list]

```
ok
```

Testing feature 4's acceptance checks 1–6 in the preview:

[Tool: Browser navigate — http://localhost:4173/]

```
navigated to http://localhost:4173/
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "check1": {
    "items": [
      "Smartphone ONE MODEL: 38.4 g CO2e a day · water: 11.0 L a day, mostly grey water",
      "Laptop ONE MODEL: 75.3 g CO2e a day · water: no manufacturing water figure found",
      "TV, about 55-inch PROXY: 48.3 g CO2e a day · water: no manufacturing water figure found"
    ],
    "total": 161.97
  },
  "check2_phone2": "Smartphone × 2 ONE MODEL: 76.7 g CO2e a d ... [truncated]"
}
```

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "check7_chart": [
    "A new 55-inch TV (manufacturing) 370 kg CO2e",
    "A new laptop (manufacturing) 110 kg CO2e",
    "A new smartphone (manufacturing) 42 kg CO2e"
  ],
  "check7_links": [
    "Samsung signage proxy -> https://www.samsung.com/global/sustainability/landing_hub-file/AY_wFeS6BzIALYNu/Signage_QMC_Environmental_Report_EN_2503.pdf",
    "Apple MacBook Air -> https://www.apple.co ... [truncated]"
  ]
}
```

Check 7 passes, and in the combined run energy stays at 366.4 Wh while carbon rises by exactly 162 g. Checking the layout:

[Tool: Browser find — My devices]

Found 1 match(es) for "My devices":
– generic "My devices" [ref_285]

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — scroll_to]

scrolled ref_285 into view

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x609

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — scroll]

scrolled down at (400, 300)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 800x600

Tab Context:

- Executed on tabId: seed

- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Looks right. Checking phone width in dark mode:

[Tool: Browser resize_window — mobile]

Viewport set to 375x812 (mobile) on this tab (scaled down to fit if larger than the pane). Reset it with preset "desktop" as soon as you finish testing; the desktop app also clears a size you set when your turn ends or the pane's width changes, so set it again if you still need it. If the user wants to keep a phone or tablet size after your turn, they can pick Mobile or Tablet in the pane's Viewpo ... [truncated]

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "iw": 375,
  "overflow": false,
  "sw": 375,
  "y": 3480.6640625
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

– Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

Works on a phone. Resetting the viewport, running a regression pass on features 1–3, and clearing test data:

[Tool: Browser resize_window — desktop]

Viewport emulation cleared; the tab is back to the pane's responsive size (desktop). Color scheme emulation set to light on this tab; it survives reloads until you set the other value or the pane re-syncs the tab to the app theme.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "digitalCarbonWithDevices": 301.21,
  "estimate": "708 kWh (74.1 kWh–6,246 kWh) · 271 kg CO2e (30.0 kg CO2e–2.38 t CO2e) · 3,352 L (259 L–30,376 L)",
  "image": [
    [
      1.35,
      0.086,
      11.49
    ],
    [
      0.513,
      0.0327,
      4.3662
    ],
    [
      0.0066,
      0.0004,
      0.0561
    ]
  ],
  "session": [
    [
      46.652,
      17.686,
      238.219
    ],
    ... [truncated]
  ]
}
```

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Regression passes. Recording the S51 decision in the spec and updating the plan:

[Tool: Bash — Record S51 decision; tick checkpoint 5 tasks]

```
M assets/css/features.css
M assets/js/app.js
M assets/js/data.js
M assets/js/digital.js
M index.html
M plan.md
M spec.md
?? assets/js/devices.js
```

Checkpoint 4 is committed as c6624c1. Checkpoint 5 (device manufacturing) is built, and all seven of feature 4's acceptance checks pass. Checkpoint 5 isn't committed yet.

Sources: I opened all 12 links and found the figures in the source text: iPhone 17 at 55 kg (53% + 23% production), MacBook Air at 155 kg (71%), 75Q60C at 982 kg (70.4%), signage display at 1,691 kg (21.9%), OptiPlex at 198 kg, Xbox at 190 kg production, 112 kg ("lower end"), P2725H at 194 kg, S2721H at 523 kg (63.9%), the PS4 at 89 kg, and devera.ai at 215.1 kg. As you decided, S51's mismatched litre figure is disclosed and the gallons figure is used.

What changed - Data ([data.js](#)): the eight device figures, each with its default years, label (one model, proxy, whole life cycle, stand-in or lower bound), basis and source link. The phone also has its production water. - **"My devices" list** (new [devices.js](#), inside the digital-day panel): each row has a device type, number owned, years kept and people sharing. Years reset to the source's default when you change the type, and the list is saved in your browser. - **Manufacturing component:** - It's added to the digital-day total in carbon, and in water for phones only. - The breakdown lists each device, and every device other than the phone reads "no manufacturing water figure found". - The digital-day card says energy excludes manufacturing and that the phone's water is mostly grey water, measured differently. Nothing is added to the AI total. - **Yearly comparisons:** smartphone 70 → 42 kg, laptop 250 → 110 kg and TV 350 → 370 kg, now labelled "(manufacturing)". The methodology text links S39, S40 and S43 in place of the old Apple home page, devera.ai and the EPA document. I left the smartphone's *water* comparison for feature 5, as the spec says. - **Methodology:** a new "Device manufacturing" paragraph, plus a limitations note with links to all 12 sources.

Acceptance checks

#	Result
1	162.0 g a day: phone 38.4, laptop 75.3, TV 48.3 ✓
2	TV not shared: 144.8 g (displayed "145 g" by the page's rounding); 2 phones: 76.7 g ✓
3	Phone kept 6 years: 19.2 g ✓

#	Result
4	Water 11.03 L a day, phone only, "mostly grey water"; the others show "no manufacturing water figure found" ✓
5	Digital-day energy excludes manufacturing (0 Wh with devices only; still 366.4 Wh with feature 3's typical day), with the note beside it ✓
6	Changing devices leaves the AI total, budget and team view unchanged ✓
7	Yearly chart shows 42, 110 and 370 kg with links to S39, S40 and S43 ✓

With feature 3's typical day added, carbon is $139.2 + 162.0 = 301.2$ g exactly. Features 1–3 still pass their checks. The layout works at phone width and in dark mode, with no console errors, and I cleared the test data.

In `plan.md` I ticked the agent tasks and added a Revisions entry for the S51 decision; the same decision is recorded in `spec.md`. "User verifies checkpoint 5" is left for you. Should I commit checkpoint 5 and start checkpoint 6 (like-for-like comparisons)?

10. Learner (RG Sureshkumar)

commit checkpoint 5 and start checkpoint 6

Agent

[Tool: Bash — Commit checkpoint 5; read feature 5 sources]

```
e793423 Add device manufacturing to the digital day (feature 4)
### S50. Water Footprint Network, "What is a water footprint?"

- Citation: Water Footprint Network. What is a water footprint?
<https://www.waterfootprint.org/water-footprint-2/what-is-a-water-footprint/>
- Claim or figure: definitions of green, blue, and grey water footprints; one kilogram
of beef requires about 15,000 litres (93% g ... [truncated]
```

Checkpoint 5 is committed as e793423. Checkpoint 6 changes a lot of figures, so I'll first check its sources and values directly:

[Tool: Bash — Check feature 5 source URLs]

```
200 s67 https://hess.copernicus.org/articles/15/1577/2011/hess-15-1577-2011.pdf
200 s67doi https://doi.org/10.5194/hess-15-1577-2011
200 s68 https://waterfootprint.org/media/downloads/Mekonnen-Hoekstra-2012-
WaterFootprintFarmAnimalProducts.pdf
200 s68doi https://doi.org/10.1007/s10021-011-9517-8
200 s69 https://www.eia.gov/tools/faqs/faq.php?id=97
200 s70 https://waterfootprint.org/resources/Repor ... [truncated]
```

Two DOIs return 403 (Science and Wiley block automated requests), so I'll confirm those through Crossref. Now the key figures in each source:

[Tool: Bash — Verify key feature 5 figures in sources]

```
Reducing food's environmental impacts through producers and consumers | Science |  
https://www.science.org/doi/10.1126/science.aaq0216  
Carbon Footprint Assessment of a Paperback Book | Journal of Industrial Ecology |  
https://onlinelibrary.wiley.com/doi/10.1111/j.1530-9290.2011.00414.x  
Beef (beef herd),2010,99.48  
Beef (dairy herd),2010,33.3  
Coffee,2010,28.53  
== s67 88133  
    Coffee, roasted.{0,120} - ... [truncated]
```

Most figures confirmed. Two gaps remain: I need the egg row in S68, and the IOP pages for S75/S76 returned only a short placeholder page. Trying their PDFs:

[Tool: Bash — Check egg figure and IOP paper PDFs]

```
200 text/html; charset=UTF-8  
200 text/html; charset=UTF-8  
egg -> Egg Grazing 3,952 375 1,189 10,604 1,360 1,176 1,695 76 161 1,740 183 331 6,781  
418 446 7,644 Mixed 2,351 230 708 6,309 815 699 1,085 51103  
egg -> Eggs 2,592 244 429 3,265 1,425 111 100 2.29 29 33 Chicken meat 3,545 313 467  
4,325 1,440 127 100 3.00 34 43 Butter 4,695 465 393 5,553 7,692  
== s75.pdf None  
== s76.pdf None  
Shell cwd was ... [truncated]
```

Eggs confirmed at 244 L/kg. IOP serves automated clients a placeholder page, so I'll read S75 and S76 in the browser pane instead:

[Tool: Browser tabs_create]

```
{  
  "serverId": "preview-local_75563973-ee26-49cb-87d9-d976bbd0b65c",  
  "tabId": "tab-1",  
  "reused": false,  
  "type": "browser"  
}  
Opened tab tab-1 in the background – the user's current tab stays in front. Use  
`navigate` with tabId "tab-1" to load a URL; front it with `tabs_select` when the user  
should look.
```

[Tool: Browser navigate — <https://iopscience.iop.org/article/10.1088/1748-9326/aa7541>]

```
navigated to https://iopscience.iop.org/article/10.1088/1748-9326/aa7541
```

Tab Context:

- Executed on tabId: tab-1
- Available tabs:
 - tabId tab-1: "The climate mitigation gap: education and government recommendations"

```
miss the most effective individual actions – IOPscience"
("https://iopscience.iop.org/article/10.1088/1748-9326/aa7541")
```

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "carfree": "r child (an average for developed countries of 58.6 tonnes CO2-equivalent
(tCO2e) emission reductions per year), living car-free (2.4 tCO2e saved per year),
avoiding airplane travel (1.6 tCO2e saved per round",
  "dry": " <0.2 tCO2e). Compare for instance two actions cited as among the most
effective ways to reduce household energy usage (hang drying clothing (0.21 tCO2e) and
wash ... [truncated]
```

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "r021": [
    " as among the most effective ways to reduce household energy usage (hang drying
clothing (0.21 tCO2e) and washing clothing in cold water (0.25 t"
  ],
  "recycle": [
    "reater potential to reduce emissions than commonly promoted strategies like
comprehensive recycling (four times less effective than a plant-based ",
    "were framed in such a way that they would take the maxim ... [truncated]
```

[Tool: Browser navigate — https://iopscience.iop.org/article/10.1088/1748-9326/ab8589]

```
navigated to https://iopscience.iop.org/article/10.1088/1748-9326/ab8589
```

Tab Context:

- Executed on tabId: tab-1
- Available tabs:
 - tabId tab-1: "Radware Bot Manager Captcha" ("https://validate.perfdribe.com")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "bev": [],
  "carfree": [],
  "heat": [],
  "hybrid": [],
  "len": 358,
  "renew": [],
  "renov": [],
  "vegan": []
}
```

```
(captured at origin https://validate.perfdribe.com)
```

Tab Context:

- Executed on tabId: tab-1
- Available tabs:
 - tabId tab-1: "Radware Bot Manager Captcha" ("https://validate.perfdrive.com/fb803c746e9148689b3984a31fccd902/?ssa=5c969125-a9b3-42ad-b12d-5f2a6e2ca9a0&ssb=780722 ... [truncated]

S76's page shows a bot-check CAPTCHA, which I won't try to get past. Its figures rest on the research-stage check, and I'll flag that. Closing the tab:

[Tool: Browser tabs_close]

Closed tab tab-1.

Now the current methodology text that this rewrite covers:

[Tool: Bash — Print methodology from Your footprint onward]

```
<strong>Your footprint.</strong> "Where you live" is a regional baseline for goods,
    services, and shared infrastructure (per-capita consumption from
    <a href="https://ourworldindata.org/co2-emissions" target="_blank"
rel="noopener">Our World in Data</a>);
    home energy and driving are separate so they do not double-count.
    <strong>Home</strong> runs from a small apartme ... [truncated]
```

Next, where the comparison lists and itemBase are used across the code:

[Tool: Bash — Find comparison-item usages and bar CSS]

```
assets/js/core.js:15:      SIZE_LABEL_OVERRIDES, GAL_TO_L, DAYS } = AIPF;
assets/js/core.js:235:    return ((loc.w + home.w + drive.w + diet.w + fly.w) *
GAL_TO_L) / DAYS;
assets/js/core.js:237:  function itemBase(item, metric) { return metric === 'carbon' ?
item.c * 1000 : item.w * GAL_TO_L; }
assets/js/core.js:248:                                itemBase, dailyWords });
assets/js/core.js:273:    t ... [truncated]
```

team.js makes its own comparisons too, so I'll check those:

[Tool: Bash — Read team view comparisons]

```
function renderCompanyTotal() {
  const metric = AIPF.state.metric;
  const co = companyPerPerson(metric);
  const days = data.company.workingDays || 250;
  const yr = co.dailyTotal * days;

  root.querySelector('#aipf-tm-co-daily').textContent =
AIPF.fmtMetric(co.dailyTotal);
  root.querySelector('#aipf-tm-co-annual').textContent = AIPF.fmtMetric(yr);
  root.querySelector('#aipf-tm- ... [truncated]
```

[Tool: Bash — Replace comparison lists and driving figures]

```
assets/js/data.js:77:    gridWaterLPerKwh: 4.35,                // Berkeley Lab:
water to generate US electricity overall
assets/js/digital.js:183:    (whElec[i] / 1000) * D.gridWaterLPerKwh + (whDc[i] / 1000)
* AIPF.DC_WATER_L_PER_KWH);
```

Removing the duplicate water-factor constant I just added, and updating `itemBase` in `core.js` to the new item shape:

[Tool: Bash — Update itemBase for new comparison items]

```
ok
```

Now the chart code in `calculator.js`:

[Tool: Read — calculator.js]

```
213   }
214
215   // ---- Bar charts ----
216   function renderBars(targetId, focalLabel, focalVal, items, unit) {
217     const rows = [{ label: focalLabel, v: focalVal, focal: true }];
218     for (const it of items) {
219       const v = AIPF.itemBase(it, state.metric);
220       if (v <= 0) continue; // skip items with no impact in this metric (keeps the
water view clean)
221       rows.push( ... [truncated]
```

[Tool: Bash — Rewrite comparison bar rendering with two focal bars]

```
ok
```

Now the cited report: the same items and both totals, plus updated references.

[Tool: Bash — Locate report chart sections]

```
343:    const dailyRows = [{ label: 'My daily AI use', v: aiDay, focal: true
}].concat(DAILY_ITEMS.map((it) => ({ label: it.label, v: it.c * 1000 })));
344:    const addRows = [{ label: 'A year of my AI use', v: aiYr, focal: true
}].concat(ANNUAL_ITEMS.filter((it) => it.dir === 'add').map((it) => ({ label: it.label,
v: it.c * 1000 })));
345:    const cutRows = [{ label: 'Cut my AI use', v: aiYr, f ... [truncated]
```

[Tool: Bash — Read report water paragraph and setup lines]

```
'<p>The same prompts consume about <strong>' + AIPF.fmtWater(wDay) + ' of water a
day</strong> (range ' + AIPF.fmtWater(wMin) + ' to ' + AIPF.fmtWater(wMax) + '), or
<strong>' + AIPF.fmtWater(wYr) + ' a year</strong> – <span class="big">' + pctWStr +
```

```
'</span> of my daily blue-water footprint (' + AIPF.fmtWater(dayW) + '). All water figures here are blue water (freshwater drawn from rivers, lakes, ... [truncated]
```

[Tool: Bash — Update report charts and references]

```
13:          DEFAULT_ROWS, REPORT_REFS, GAL_TO_L, DAYS, WPM, LEGACY_SIZE_IDS } = AIPF;
```

References 17–25 line up with the footnote numbers. Removing the now-unused GAL_TO_L import, and adding CSS for the item notes:

[Tool: Bash — Drop unused import; style comparison notes]

```
ok
```

Now the methodology rewrite. First the "Your footprint" paragraph (driving and the burger water sentence):

[Tool: Bash — Rewrite driving and baseline water text]

```
ok
```

Now replacing the four comparison paragraphs with text that matches the new figures:

[Tool: Bash — Rewrite comparison methodology; check removed refs]

```
(Bash completed with no output)
```

All the removed items' old citations are gone from the page. Now checking feature 5's acceptance checks in the preview:

[Tool: Browser navigate — http://localhost:4173/]

```
navigated to http://localhost:4173/
```

```
Tab Context:
```

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

```
No console logs.
```

```
Tab Context:
```

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "allItemsHaveNoteAndLink": true,
  "carbonAdd": [
    "A year of driving = 4,600 kg CO2e",
    "A transatlantic round-trip flight = 1,600 kg CO2e",
    "A beef burger every week = 585 kg CO2e",
    "A new 55-inch TV (manufacturing) = 370 kg CO2e",
    "A new laptop (manufacturing) = 110 kg CO2e",
    "A year of daily coffee = 73 kg CO2e",
    "A new smartphone (manufacturing) = 42 kg CO2e", ... [truncated]
```

[Tool: Bash — Link second sources; scan for full-footprint figures]

(Bash completed with no output)

No full-footprint figures remain apart from the explained 15,400 L/kg. Now check 1's second half: no removed item left in the data or the methodology:

[Tool: Bash — Search for removed comparison items]

microwave	data:0	html:0
shower	data:0	html:0
dishwasher	data:0	html:0
dryer	data:0	html:0
air conditioning	data:0	html:1
T-shirt	data:2	html:1
sofa	data:0	html:0
bicycle	data:0	html:0
new car	data:0	html:0
LED	data:7	html:16
short-haul	data:0	html:0
cross-country	data:0	... [truncated]

[Tool: Bash — Remove the removed-items sentence]

systems that rely on irrigated feed (Hess & Williams 2023), so the burger is shown

with that range (about 8–226 L). A day of home electricity, about 129 L, is the US

average of 10,791 L ... [truncated]

Now the report, mobile layout, and regression checks with a digital day entered:

[Tool: Browser navigate — http://localhost:4173/]

navigated to http://localhost:4173/

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "dailySub": "A day of your AI use is 21 g CO2e and your digital day is 139 g CO2e (both highlighted, shown separately).",
  "f3": [
    366.42,
    347.57,
    385.27
  ],
  "focalAdd": [
    "A year of your digital day 51 kg CO2e",
    "A year of your AI use 7.6 kg CO2e"
  ],
  "focalDaily": [
    "Your digital day 139 g CO2e",
    "Your daily AI use 21 g CO2e"
  ],
  "reportDigitalBars": 6, ... [truncated]
```

The report and focal bars work. twoLinks came back empty only because the carbon view was showing, and both two-source items are water items. Checking the water view at phone width in dark mode:

[Tool: Browser resize_window — mobile]

Viewport set to 375x812 (mobile) on this tab (scaled down to fit if larger than the pane). Reset it with preset "desktop" as soon as you finish testing; the desktop app also clears a size you set when your turn ends or the pane's width changes, so set it again if you still need it. If the user wants to keep a phone or tablet size after your turn, they can pick Mobile or Tablet in the pane's Viewpo ... [truncated]

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "overflow": false,
  "two": [
    "US EIA + Berkeley Lab",
    "Mekonnen & Hoekstra 2012 + Hess & Williams 2023"
  ]
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

```
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when yo ... [truncated]
```

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — scroll]

scrolled down at (375, 800)

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

[Tool: Browser computer — screenshot]

[screenshot]

Screenshot size: 450x974 0.6-scale view; coordinate frame: 750x1624.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")
- Viewport: emulating 375x812 (you set this; reset it with preset "desktop" when you finish testing; the app also clears it when your turn ends or the pane's width changes)

The layout works. But the text under the chart says the calculator "Excludes training, image and video generation...", and images are now included. Finding that text:

[Tool: Bash — Find the outdated exclusions note]

```
index.html:31:         average, plus the hardware's embodied emissions. Training,
index.html:371:         <p class="aipf-fine">Excludes training, image and video
generation, and retries.</p>
```

[Tool: Bash — Read the intro scope note]

```
<a href="https://huggingface.co/spaces/genai-impact/ecologits-calculator"
target="_blank" rel="noopener">EcoLogits</a>,
    an open-source estimator of the energy, carbon, and water behind each AI model.
</p>
<details class="aipf-dek-more">
    <summary>What's counted, and how reliable it is</summary>
    <p>
        Carbon uses the average grid intensity of the region you pick ... [truncated]
```

Both notes predate feature 1. Correcting them, since generated images are now counted when logged:

[Tool: Bash — Correct the outdated exclusion notes]

ok

Running a final regression pass over features 1–4 on the finished page:

[Tool: Browser resize_window — desktop]

Viewport emulation cleared; the tab is back to the pane's responsive size (desktop).
Color scheme emulation set to light on this tab; it survives reloads until you set the
other value or the pane re-syncs the tab to the app theme.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

"ok"

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
javascript_tool failed: ReferenceError: AIPF is not defined
    at <anonymous>:1:12
```

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "f1_image": [
    [
      1.35,
      0.086,
      11.49
    ],
    [
      0.513,
      0.0327,
      4.3662
    ],
    [
      0.0066,
      0.0004,
      0.0561
    ]
  ],
  "f2_estimate": "708 kWh (74.1 kWh–6,246 kWh) · 271 kg CO2e (30.0 kg CO2e–2.38 t CO2e) · 3,352 L (259 L–30,376 L)",
  "f2_session": [
    [
      46.652,
      17.686,
      238.219
    ],
    [
      18.178,
      7.171 ... [truncated]
    ]
  ]
}
```

Every earlier feature still matches its acceptance values, with no console errors. Updating the plan:

[Tool: Bash — Tick checkpoint 6 agent tasks in the plan]

```
M assets/css/base.css
M assets/js/calculator.js
M assets/js/core.js
M assets/js/data.js
M index.html
M plan.md
```

Checkpoint 5 is committed as e793423, and checkpoint 6 (like-for-like comparisons) is built. All six of feature 5's acceptance checks pass. Checkpoint 6 isn't committed yet, and one question is left for you (below).

Sources

Almost every feature 5 figure matches its source. I opened the pages and found: - the Mekonnen & Hoekstra table rows (139, 301, 443, 283 and 3,816 L/kg; beef 550, eggs 244, milk 86) - cotton at 4,917 L/kg and EIA's 10,791 kWh - the Our World in Data table for Poore & Nemecek (beef 99.48 and 33.3 kg/kg, coffee 28.53 kg/kg) - the EPA's 11,500 miles and 4.6 t, and Levi's 33.4 kg - Hess & Williams' 67 L/kg and 2,000 L/kg - Wynes & Nicholas' 2.4 t, 1.6 t, 0.21 t (hang-drying) and 1.15 t

Three limits: - **Ivanova et al. (S76)**: the page shows a bot-check CAPTCHA, which I didn't try to get past. The hybrid, heat-pump, retrofit, vegan, green-electricity and electric-car figures rest on the research-stage check. You may want to open that page yourself. - **Wynes & Nicholas recycling (0.21 t)**: it appears only in a figure. The text's "four times less effective than a plant-based diet" ($0.8 \div 4 = 0.2$ t) is consistent with it. - **Two DOIs (Poore & Nemecek in Science, Wells in Wiley)** block automated requests. Crossref confirms both, and the chart links Our World in Data's page for the Poore figures.

What changed - Data (data.js): the four comparison lists now hold exactly the spec's items and values. Each item has its assumption and boundary, source link and reported range. PS5 carbon and home-electricity water are calculated from kWh. - **Driving setting**: about 1,150, 4,600 and 9,580 kg. The option labels say "scaled from the EPA average" or "EPA, 11,500 miles". - **Charts (calculator.js)**: - Your AI total and your digital day are two separate highlighted bars in the daily, yearly-add and yearly-cut charts. When no digital day is entered, that bar reads "(not entered yet)". - Every item shows a note line such as "113 g patty · farm to retail; low end is dairy-herd beef · range ... · source". - **Report**: the same items, both totals, and 9 new references (25 in total). - **Methodology**: I rewrote the comparison paragraphs to match every figure. They explain the chosen rule and why beef's 15,400 L/kg total isn't comparable (only 550 L/kg of it is blue water). All citations for removed items are gone. - **Two notes fixed**: both still said image generation was excluded, which stopped being true at checkpoint 2.

Acceptance checks

#	Result
1	Every item matches the spec's tables; no removed item remains in the data, methodology or report ✓
2	Every item has a working source link and a visible boundary or assumption; the two items with two sources link both ✓
3	Water view: coffee 0.97 L, burger 62 L (7.6–226 L), home electricity 129 L ✓
4	PS5 hour: 82 g in the US and 27 g in the UK ✓
5	Both totals shown as separate highlighted bars in all three charts; neither is shown as a share of the other ✓
6	No full-footprint water figure is presented as comparable ✓

I re-ran features 1–4 on the finished page; all their figures are unchanged. The layout works at phone width and in dark mode, with no console errors, and I cleared the test data. In plan.md I ticked the checkpoint 6 agent tasks and the two "across all checkpoints" items.

Decision needed. The team view's company card compares yearly AI water to "ten-minute showers", using EPA WaterSense's 21 gallons. That's water withdrawn rather than consumed, and it's not in the spec's tables, although the spec's rule would remove it. Should I: - **remove it**, and add a revision note to the spec, or - **leave it**, and note it as out of scope?

After that, tell me whether to commit checkpoint 6. Once it's committed, all implementation tasks are done. What's left is yours: the four Verification items in `plan.md` (including assessing sources S59–S80) and the Delivery items.

This chat's transcript is a deliverable too. When you're ready, say `save transcript` and I'll save it as `transcripts/build-2026-09-26_HHMMSS.md`.

11. Learner (RG Sureshkumar)

remove the shower comparison and commit checkpoint 6

Agent

[Tool: Bash — Remove shower anchor; record revision]

```
(Bash completed with no output)
```

Verifying the team view in both metrics:

[Tool: Browser javascript_tool — javascript_exec]

```
"ok"
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "carbon": "",
  "teamVisible": false,
  "water": ""
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

```
No console logs.
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Both anchors came back empty because the team panel is hidden. I'll check when it renders so I can test it properly:

[Tool: Bash — Find team panel visibility logic]

```
110: function renderCompanyTotal() {
193:   renderCompare(); renderCompanyTotal(); renderTeamBars(); renderIfEveryone();
210:   if (panel) panel.hidden = false;
219:   if (!applyData(AIPF.TEAM_DATA)) { panel.hidden = true; return; }
220:   panel.hidden = false;
224:   if (warn) warn.hidden = !data.sample;
```

[Tool: Bash — Read team init logic]

```
// ---- Wiring ----
function applyData(next) {
  data = next;
  if (!data || !Array.isArray(data.teams) || !data.teams.length) { data = null;
return false; }
  const saved = AIPF.store.get('team:mine', null);
  team.myTeam = data.teams.some((t) => t.id === saved) ? saved : data.teams[0].id;
  return true;
}

// Public hook so a deployment can supply real roster data after it loads. ...
[truncated]
```

The team panel should be visible. Earlier in this session its contents read fine, so something may have broken it. Checking:

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "anchor": "About 3.4 transatlantic round-trip flights a year.",
  "cells": "YOU, TODAY\n\n20.8 g CO2e\n\n27 prompts\n\nENGINEERING AVERAGE\n\n17.2 g CO2e\n\n37 prompt",
  "daily": "22.0 kg CO2e",
  "hasInit": "function",
  "hidden": false,
  "ready": "1",
  "teams": 8
}
```

Tab Context:

- Executed on tabId: seed

- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http ... [truncated]

That earlier read ran before the page had finished loading. The panel is fine, and the carbon anchor shows "About 3.4 transatlantic round-trip flights a year." Now the water view:

[Tool: Browser javascript_tool — javascript_exec]

```
{
  "carbonAgain": "About 3.4 transatlantic round-trip flights a year.",
  "water": {
    "anchor": "",
    "daily": "200 L"
  }
}
```

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

[Tool: Browser read_console_messages]

No console logs.

Tab Context:

- Executed on tabId: seed
- Available tabs:
 - tabId seed: "AI Footprint Calculator – Company Edition" ("http://localhost:4173/")

Water view has no anchor now; carbon keeps its flight anchor. Committing checkpoint 6:

[Tool: Bash — Commit checkpoint 6 on project-2]

```
87fdd88 Make everyday comparisons like for like (feature 5)
e793423 Add device manufacturing to the digital day (feature 4)
c6624c1 Add the digital day builder (feature 3)
3210da9 Add agent sessions and projects (feature 2)
397e7eb Add image generation logging (feature 1)
6b57f78 Compute the AI total in one place and give log entries a kind
ba11478 Add Project 2 implementation plan and planning-st ... [truncated]
```

I removed the shower comparison and committed checkpoint 6 as 87fdd88 on project-2. Nothing is pushed, and main is untouched.

The removal - In the water view, the team view's company card now shows no comparison. The carbon view keeps "About 3.4 transatlantic round-trip flights a year." - I checked both views in the preview, with no console errors. - The decision is recorded as a revision in spec.md and in the plan (Revisions, plus a ticked task).

Implementation is done: all six checkpoints are built and committed.

Commit	Checkpoint
6b57f78	1: one shared AI total, log entry kinds
397e7eb	2: image logging
3210da9	3: agent sessions and projects
c6624c1	4: digital day builder
e793423	5: device manufacturing
87fdd88	6: like-for-like comparisons

Still yours in p1an.md: - the six "User verifies checkpoint N" items - the four Verification items, including assessing sources S59–S80 in research.md; Ivanova et al. (S76) is worth opening yourself, because I couldn't re-read it this session - the Delivery items: exporting transcripts, then the debrief

This chat's transcript is a deliverable. Say `save transcript` and I'll save it as `transcripts/build-2026-09-26_HHMMSS.md`.

12. Learner (RG Sureshkumar)

save transcript

Agent

[Tool: Bash — Read transcript conventions and prior example]

```
# Transcripts

Session transcripts for this project, kept as evidence of process.

- `session-2026-09-12.md` - the build session: porting Andy Masley's calculator
  to a standalone site and adding the three features.
- `debrief-*.md` - Unit 1 debrief conversations, saved separately so debrief
  evidence stays distinct from build sessions.
- `debrief-*.pdf` - the same debrief rendered for submissio ... [truncated]
```

[Tool: Bash — Locate session log and PDF tools]

```
-rw-----@ 1 <user>  staff   606307 Sep 26 15:31 <session-log>
-rw-----@ 1 <user>  staff   3990659 Sep 26 18:26 <session-log> ... [truncated]
```

[Tool: Bash — Read PDF memory; check browsers and log shape]

```
---
name: transcript-export-as-pdf
description: "The learner wants session transcripts exported as PDF, not only as
Markdown."
```

```
metadata:
  node_type: memory
  pinned: false
  originSessionId: 9922e71d-ff53-4e5e-bd44-49c6ad7c09ad
  modified: 2026-09-12T05:09:00.458Z
---
```

Transcripts are wanted as PDF

Across two separate sessions in this course project, the learner has asked for the session tran ... [truncated]

[Tool: Bash — Inspect session log message structure]

```
USER STR None sdk 'start implementation'
USER STR None sdk 'start checkpoint 1 now'
USER STR None sdk 'commit checkpoint 1'
USER STR None sdk 'start checkpoint 2'
USER STR None sdk 'go with b and then commit checkpoint 2'
USER STR None sdk 'start checkpoint 3'
USER STR None sdk 'the side effects are ok and you commit checkpoint 3 and then start
checkpoint 4'
USER STR None sdk 'commit checkpoint 4 ... [truncated]
```

[Tool: Write — make_transcript.py]

File created successfully at: <scratchpad> (file state is current in your context – no need to Read it back)

Quick status: I'm generating the Markdown transcript from this session's log. Then I'll render the matching PDF and update transcripts/README.md.

[Tool: Bash — Generate and sanity-check the build transcript]

```
12 learner turns; 136107 chars
transcripts/build-2026-09-26_230322.md
0
29:## 1. Learner (RG Sureshkumar)
33:## Agent
87:## Instructions for the user
170:## 2. Learner (RG Sureshkumar)
174:## Agent
628:## 3. Learner (RG Sureshkumar)
632:## Agent
653:## 4. Learner (RG Sureshkumar)
657:## Agent
1236:## 5. Learner (RG Sureshkumar)
1240:## Agent
1358:## 6. Learner (RG Sureshkumar)
1362:## Agent
2237:# ... [truncated]
```

Two things to check: a stray heading at line 87, and turns 8 and 9 (the duplicate message) coming out as separate turns.

[Tool: Bash — Inspect the two suspect spots]

```
Exit code 1
'''text
# Implementation Plan

> EDITING DIRECTIVE: USER AND AGENT EDIT THIS FILE COLLABORATIVELY. THE USER MUST
REVIEW AND APPROVE ITS CONTENT.

Purpose of this file: Turn the approved specification into ordered, updatable
implementation and verification work.

## Instructions for the user

Preserve the approved requirements and verify the completed work. Direct priorities,
scope, and ... [truncated]
```

Line 87 is inside a code block, so it's fine. Now the duplicate-message turn:

[Tool: Bash — Inspect turns 8 and 9]

```
- The typical day is saved in your browser, so it persists between visits.

"User verifies checkpoint 4" is still unticked in `plan.md`. Should I commit checkpoint
4 and start checkpoint 5 (device manufacturing)?

## 8. Learner (RG Sureshkumar)

commit checkpoint 4 and start checkpoint 5

## Agent

You've hit your session limit · resets 10:40pm (America/Chicago)

## 9. Learner (RG Sureshkumar)

c ... [truncated]
```

That's accurate: the first message hit a session limit, and you sent it again. I'll leave it as is. Now rendering the PDF through headless Chrome, as for the earlier transcripts:

[Tool: Bash — Render the transcript PDF with headless Chrome]

```
[64646:10182806:0926/230345.499331:ERROR:ui/display/mac/cv_display_link_mac.mm:195]
CVDisplayLinkCreateWithCGDisplay failed. CVReturn: -6670
708306 bytes written to file ~/Documents/College/Undergraduate School/Semesters/Fall
2026/Academics/AI Explorations/Unit 1 Brief - AI Footprint
Calculator/transcripts/build-2026-09-26_230322.pdf
-rw-r--r--@ 1 <user> staff 137704 ... [truncated]
```

Now the README entry for the new kind of transcript:

[Tool: Edit — README.md]

The file ~/Documents/College/Undergraduate School/Semesters/Fall 2026/Academics/AI Explorations/Unit 1 Brief – AI Footprint Calculator/transcripts/README.md has been updated successfully.

Checking that your multiple-choice answers came through in full:

[Tool: Bash — Verify decision answers appear untruncated]

```
2334:_[Tool: AskUserQuestion – Console W / Router mix / Log window]_
2335-
2336-'''text
2337-Your questions have been answered: "The spec gives the console's streaming/menus
draw only as a range, 43–47 W (S33: 43.4 W and 47.06 W on the PS5 home menu), with no
central value. Which centr
--
2858:_[Tool: AskUserQuestion – Phone water]_
2859-
2860-'''text
2861-Your questions have been answered: "S51 s ... [truncated]
```

I saved this chat's transcript in two formats:

- [transcripts/build-2026-09-26_230322.md](#)
- [transcripts/build-2026-09-26_230322.pdf](#), 78 pages, rendered from the Markdown so the two match

Format: the same as the research, spec and plan transcripts. - Your 12 turns are headed "Learner (RG Sureshkumar)"; my replies are under "Agent". - Each tool call is a one-line summary, with its output cut to 400 characters. Screenshots appear as [screenshot], and my internal reasoning isn't included. - Your answers to my two multiple-choice questions (the feature 3 details and the phone-water figure) are shown in full, because they record decisions. The note at the top of the file says so. - Turn 8 shows the session-limit message, and turn 9 your repeated "commit checkpoint 4 and start checkpoint 5". That's how it happened, so I left it as is.

Redaction: following transcripts/README.md, the email address, home path, username and temporary paths are replaced with placeholders. I checked the file afterwards and none remain.

README: I added an entry for the new build-* transcripts to transcripts/README.md.

The two transcript files and the README change aren't committed. Tell me when you'd like them committed.

13. Learner (RG Sureshkumar)

commit the transcript

Agent

[Tool: Bash — Commit the build transcript on project-2]

```
project-2
ab1a9a0 Add Project 2 implementation-stage transcript
```

I committed the transcript on project-2 as ab1a9a0. The commit contains the Markdown file, the matching PDF and the updated transcripts/README.md. The working tree is clean, and nothing is pushed.

The next steps in plan.md are yours: - tick the six "User verifies checkpoint N" items once you've checked each one - work through the four Verification items, including assessing sources S59–S80 - then the debrief

14. Learner (RG Sureshkumar)

push the project-2 branch

Agent

[Tool: Bash — Check remote and unpushed commits]

```
project-2
origin https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator.git (fetch)
origin https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator.git (push)
origin/project-2
ab1a9a0 Add Project 2 implementation-stage transcript
87fdd88 Make everyday comparisons like for like (feature 5)
e793423 Add device manufacturing to the digital day (feature 4)
c6624c1 Add the digital day buil ... [truncated]
```

[Tool: Bash — Push project-2 to origin]

```
To https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator.git
ba11478..ab1a9a0 project-2 -> project-2
## project-2...origin/project-2
```

I pushed project-2 to GitHub, taking it from ba11478 to ab1a9a0. That uploaded 7 commits: the six checkpoints and the transcript. The local branch now matches GitHub.

main wasn't touched, so the GitHub Pages site still shows the submitted Project 1 version.

15. Learner (RG Sureshkumar)

tick the six user verification items

Agent

[Tool: Bash — Find the six verification items]

```
76:- [ ] User verifies checkpoint 1; commit when asked
86:- [ ] User verifies checkpoint 2; commit when asked
97:- [ ] User verifies checkpoint 3; commit when asked
107:- [ ] User verifies checkpoint 4; commit when asked
117:- [ ] User verifies checkpoint 5; commit when asked
128:- [ ] User verifies checkpoint 6; commit when asked
```

[Tool: Bash — Tick the six user verification items]

```
76:- [x] User verifies checkpoint 1; commit when asked. Ticked at the user's direction
on 2026-09-26 ("tick the six user verification items"); committed as `6b57f78`
86:- [x] User verifies checkpoint 2; commit when asked. Ticked at the user's direction
on 2026-09-26 ("tick the six user verification items"); committed as `397e7eb`
97:- [x] User verifies checkpoint 3; commit when asked. Ticked at th ... [truncated]
```

I ticked the six "User verifies checkpoint" items in plan.md, as you asked. Each one records that it was ticked at your direction on 2026-09-26, quotes your words, and names its commit (6b57f78 through 87fdd88).

The change isn't committed or pushed yet. Tell me when you want either.

The four items under **Verification** are still unticked; they're yours to check and tick. They cover: - checking the calculations against the spec and its sources yourself - assessing sources S59–S80 in research.md - confirming every claim has a working citation - confirming the calculator runs locally and serves all three reference profiles

16. Learner (RG Sureshkumar)

commit and push the plan

Agent

[Tool: Bash — Commit plan.md and push project-2]

```
project-2
To https://github.com/rg-sureshkumar-goat/AI-Footprint-Calculator.git
ab1a9a0..5922577 project-2 -> project-2
5922577 Record the user's verification of all six checkpoints in the plan
## project-2...origin/project-2
```

I committed plan.md as 5922577 and pushed project-2, so GitHub now matches your local branch. main is still untouched.

The last save (build-2026-09-26_230322) doesn't include these final turns: the push, ticking the six items, and this commit. If you want the transcript to cover them, say `save transcript` and I'll save an updated copy.

17. Learner (RG Sureshkumar)

`save transcript`